

Production Networks and R&D Allocation

Yasutaka Koike	Koki Okumura
UNC Chapel Hill	CUHK*
TDB-CAREE	TDB-CAREE

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Abstract

This paper studies how production networks shape innovation and entry incentives. We develop a dynamic model in which new products inherit the supplier and buyer relationships accumulated by their firms. Innovating firms capture only part of the surplus their innovations generate for trading partners and do not internalize how innovation changes other firms' opportunities to form trading relationships. We show that reallocating R&D from entry toward incumbent expansion, and from younger toward older incumbents with more extensive trading relationships, raises welfare. We discipline the model using matched Japanese firm-to-firm transaction and patent data. In the estimated economy, the socially optimal allocation reduces long-run entry by 54%, and the transition to this allocation raises consumption-equivalent welfare by 2.8%.

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1 Introduction

How do production networks shape the allocation and efficiency of innovation? A semiconductor firm can produce a newly developed chip with inputs from long-standing suppliers and sell it to buyers it already serves. The returns to developing the chip therefore depend on accumulated trading relationships. A large literature studies how innovation diffuses through knowledge networks (Liu and Ma, 2026). Much less is known about how buyer-supplier relationships shape the private and social returns to R&D.

We develop a dynamic model of production networks, incumbent R&D, and entry to study how accumulated trading relationships shape the gap between private and social innovation returns. We discipline the model using matched Japanese firm-to-firm transaction and patent data. We show that, when trading relationships accumulate, a stationary decentralized equilibrium admits small, welfare-improving reallocations of R&D from entry to incumbent product-line expansion and, within incumbents, from younger to older firms. In the estimated economy, the planner reduces stationary entry by 53.5%, and the transition from the decentralized steady state to the planner allocation raises consumption-equivalent welfare by 2.78%.

We link Japanese firm-to-firm transaction network data spanning more than two decades with patent records to examine how trading relationships and innovation evolve over the firm life cycle. Three empirical facts motivate the model. First, supplier and buyer counts rise early in the firm life cycle conditional on firm characteristics. Second, older firms tend to trade with older partners. Third, firms whose trading partners file more patent applications also patent more themselves. In the joint instrumental-variable specification, which exploits partners' differential exposure to foreign patenting, a 10 percent increase in supplier or buyer patenting is associated with approximately a 0.7 or 1.7 percent increase in the focal firm's own patenting, respectively.

We build these facts into a dynamic general-equilibrium model. Firms accumulate supplier and buyer links through random meetings and relationship persistence. Incumbent R&D creates new product lines that inherit the parent firm's existing trading relationships. Under this network inheritance assumption, new products are produced with inputs from established suppliers and sold to established buyers. Entrants instead begin with a single product line and an initial set of buyer-supplier relationships. Entry and incumbent R&D draw on the same fixed aggregate supply of research labor. We quantify how accumulated trading relationships shape R&D allocation between entrants and incumbents and across incumbent firms of different ages.

Our matching and inheritance structure allows us to represent the network by age-

indexed distributions of product lines and trading relationships. This representation retains the feedback between innovation and network accumulation without tracking individual partner identities, and yields closed-form expressions for stationary trading-relationship distributions. It enables us to characterize private and social innovation returns and quantify the welfare effects of R&D reallocation.

Network accumulation changes both the returns to innovation and the ratio of private to social returns. As firms accumulate buyer relationships, a larger share of a new product line's social return comes from its use as an input by other firms. Suppliers capture only part of the associated surplus, so private returns place relatively less weight on the established buyer relationships of older firms. Product-line creation also reduces other firms' matching opportunities. The resulting current congestion cost is common across new product lines, while inherited supplier and buyer relationships raise the benefits of incumbent expansion. Together, these forces create scope for welfare-improving R&D reallocations from entry to incumbent expansion and from younger to older incumbents.

We estimate the model on Japanese manufacturing data using the level and life-cycle profile of buyer-supplier links and the average of the supplier and buyer patenting coefficients. The model also generates positive age assortativity, consistent with the untargeted partner-age profiles.

In the estimated economy, jointly correcting incomplete surplus appropriation and matching-stock congestion reduces the share of R&D devoted to entry by more than the sum of the reductions from correcting each distortion separately. A decomposition of the welfare gain from R&D reallocation attributes 66.4% to shifting R&D from entry to incumbent expansion and 33.6% to reallocating incumbent R&D across firm ages. A welfare-maximizing uniform subsidy to incumbent R&D captures 82.5% of the welfare gain from the transition to the planner allocation. The subsidy affects both margins, redirecting R&D from entry toward incumbent expansion and changing its allocation across incumbent ages through firms' equilibrium responses.

We show how production networks shape innovation incentives and the efficient allocation of R&D. Research on production networks studies shock propagation, allocative efficiency, and aggregate productivity (Acemoglu et al., 2012; Liu, 2019; Baqaee and Farhi, 2020; Bigio and La'O, 2020; Baqaee et al., 2025). We bring innovation and firm entry into this perspective by studying the returns to creating new products through accumulated supplier and buyer relationships. This connects production-network structure to the composition of innovation and its welfare consequences.

We also contribute to research on innovation, firm dynamics, and endogenous growth (Romer, 1990; Klette and Kortum, 2004; Acemoglu and Cao, 2015; Akcigit and Kerr, 2018).

Research on knowledge networks links cross-sector spillovers and endogenous network formation to innovation, growth, and the social returns to R&D (Cai and Li, 2019; Cai and Tian, 2024; Liu and Ma, 2026; Chen et al., 2026). Studies of R&D misallocation also emphasize differences between private profitability and knowledge spillovers (Ayerst, 2023) and the roles of firm heterogeneity, rents, and reallocation (Acemoglu et al., 2018; Aghion et al., 2025). A related surplus-appropriation mechanism appears in Acemoglu (2001), where firms do not internalize the surplus accruing to workers. Our contribution is to show how accumulated trading relationships shape the gap between private and social innovation returns and thereby the efficient allocation of R&D.

We develop a tractable framework for the joint evolution of production networks and innovation. Work on endogenous production networks (Oberfield, 2018; Acemoglu and Azar, 2020; Bernard et al., 2022; Eaton et al., 2026; Arkolakis et al., 2026) provides the foundation for modeling firms' trading relationships. Lim (2018) and Huneeus (2020) study how frictions in forming and adjusting firm-to-firm relationships shape production-network dynamics and the propagation of shocks. Boehm et al. (2024) show how the formation and destruction of trading relationships shape firm dynamics. Bernard (2026) studies how the joint accumulation of supplier and customer capital shapes firm size. Closest to our modeling approach are the contemporaneous contributions of Aekka and Khanna (2025) and Asai and Nirei (2026), who study the gradual accumulation of supplier and buyer relationships over the firm life cycle. Our framework makes firms' product-line expansion and their trading partners' innovation jointly determine the evolution of these networks. This joint determination lets us analyze how accumulated trading relationships shape the gap between private and social innovation returns and the efficient allocation of R&D.

Finally, our analysis relates to customer capital and investment in demand. Persistent customer relationships shape investment decisions (Gourio and Rudanko, 2014), and consumer search governs customer accumulation and market concentration (Shen, 2026). Ignaszak and Sedláček (2026) study the interaction between customer accumulation and innovation. Related work examines how advertising interacts with R&D to shape firm dynamics and growth (Cavenaile and Roldan-Blanco, 2021; Cavenaile et al., 2025). Pearce and Wu (2025) study the persistent mismatch between customer capital and productivity generated by strategic demand investment. Hubmer and Nord (2026) analyze how customer surplus and business-stealing externalities interact with pricing to shape the efficiency of demand investment. In a symmetric CES benchmark with only final-consumer relationships, inherited customer access raises private profits and social current benefits proportionally. In our model, new products also serve as inputs to other firms. As buyer

relationships accumulate, a larger share of a new product's current social return comes from intermediate-input use, while the innovator captures only part of the associated surplus.

The rest of the paper is organized as follows. Section 2 presents the data and empirical findings on the relationship between production networks, firm age, and patenting. Section 3 develops a dynamic model of production networks, incumbent R&D, and entry based on these empirical findings. Section 4 characterizes the R&D and entry wedges. Section 5 estimates the model parameters and quantifies the transition welfare decomposition, allocation counterfactuals, and policy exercises. Finally, Section 6 concludes the paper.

2 Motivating Facts on Production Networks and Innovation

Using firm-to-firm transaction data and linked patent records, we document two sets of facts that motivate the model. First, buyer-supplier networks expand early in the firm life cycle conditional on firm characteristics, and older firms tend to trade with older partners. Second, firms' own patenting is positively related to patenting by both their suppliers and their buyers. These positive relationships appear in both ordinary least squares and instrumental-variable estimates.

2.1 Data and Sample Construction

Our analysis relies on three main datasets: firm-level data provided by the TDB Center for Advanced Empirical Research on Enterprise and Economy (TDB-CAREE), the IIP Patent Database, and the OECD Triadic Patent Families (TPF) database.

We obtained firm-level data from TDB-CAREE at Hitotsubashi University. The data originate from credit research reports compiled by Teikoku Databank (TDB), a major private credit research company. The dataset contains firm characteristics such as firm age, industry classification (based on codes that closely match the Japan Standard Industrial Classification), location of headquarters, number of employees, and sales revenue. Crucially for our analysis, each firm also reports the identities of its major suppliers and buyers, allowing us to construct the production network of buyer-supplier links. The data cover both publicly listed and unlisted firms. The recorded links need not cover all of a firm's trading relationships.

The second dataset is the 2020 edition of the IIP Patent Database, developed for statistical analysis using standardized data from the Japan Patent Office (Goto and Motohashi, 2007;

Institute of Intellectual Property, 2020). Because the IIP Patent Database does not contain the firm identifiers needed for a direct merge with TDB, we first link it to the National Institute of Science and Technology Policy (NISTEP) Dictionary of Names of Companies using the corresponding IIP connection table, both version 2022_1 (NISTEP, 2022). These resources connect patent records to company names and identification numbers. Identification numbers from NISTEP then allow us to merge the IIP Patent Database with TDB. For firms without identification numbers, we supplement the match using firm names and addresses.

The third dataset is the August 2023 release of the OECD Triadic Patent Families (TPF) database (OECD, 2023), which combines patent applications filed with the European Patent Office (EPO), the Japan Patent Office (JPO), and the United States Patent and Trademark Office (USPTO) into patent families based on common priority applications. The primary data source for the TPF is the EPO’s Worldwide Patent Statistics Database, which provides harmonized and comparable information on patents from the EPO, JPO, and USPTO. We use four-character IPC codes, the year of the first JPO application, and applicant country information for each patent family.

We use different samples according to the purpose and data requirements of each analysis. For the conditional link-count age profiles, we use manufacturing firms in TDB for 1998–2021, while the partner-age profiles use firms across industries for 1998–2019. The supplementary cumulative-growth profiles follow manufacturing firms established in 1998–2004. For the partner-patenting regressions, we draw on the matched TDB-IIP source panel for 1996–2019 and use the OECD TPF data to construct the foreign-patent instruments.

2.2 Life-Cycle Network Patterns

The life-cycle evidence has two main components. Conditional on firm characteristics, supplier and buyer networks expand early in the firm life cycle, consistent with Asai and Nirei (2026). Trading partners’ ages also rise with firm age, consistent with Fujii et al. (2017).

2.2.1 Link Accumulation over the Firm Life Cycle

Figure 1 reports age coefficients for the log of one plus the number of recorded suppliers and buyers, controlling jointly for firm effects, industry-year effects, employment, capital, and sales per worker. The coefficients rise sharply from age one through the first decade and subsequently remain close to the mature-firm reference. Recorded networks are

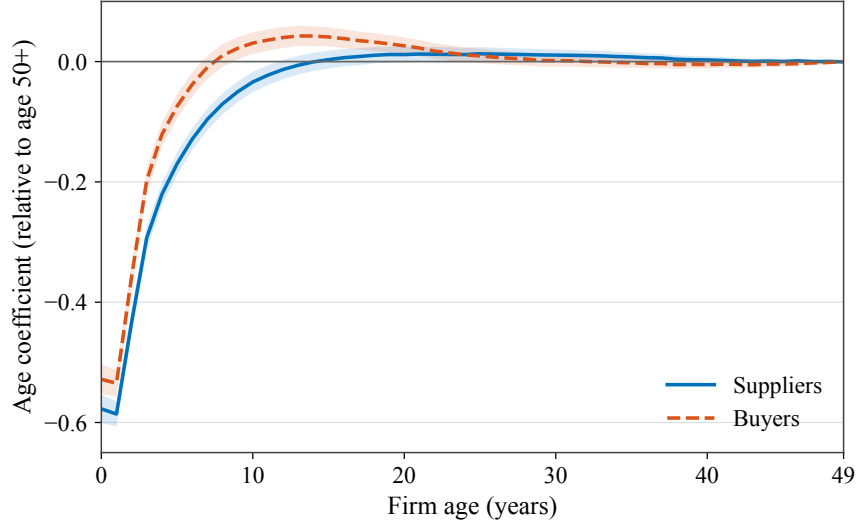


Figure 1: Trading Networks over the Early Firm Life Cycle

Notes: Age coefficients from regressions of $\log(1 + \text{distinct partners})$ on age indicators, firm and industry-year fixed effects, and log employment, log capital, and log sales per worker. Manufacturing firms, 1998–2021; 9,894,357 observations per series. Ages 50 and above form the reference group. Shading shows pointwise 95% intervals based on firm-clustered standard errors. Age is years since establishment.

initially smaller even conditional on these characteristics, with the gap narrowing early in life.

The figure describes conditional age differences rather than the total expansion associated with firm growth. Appendix A.2 details the specification and reports complementary evidence on cumulative network expansion from the initial observation near establishment.

2.2.2 Age Assortativity

Over the first fifty years, older firms generally have older trading partners, indicating positive conditional age assortativity in the cross-section. We present these partner-age profiles alongside the model’s untargted predictions in Section 5.1, Figure 3(b).

2.3 Trading-Partner Patenting and Own Patenting

Using observed firm-to-firm transaction links in the matched TDB-IIP panel, we examine whether firms patent more when their pre-existing suppliers or buyers patent more, after accounting for permanent firm differences and aggregate year shocks. The baseline regression sample contains 50,918 firm-year observations from a source panel covering 1996–2019.

Let P_{it} denote patent applications by firm i . Let $S_{i,t-1}$ and $B_{i,t-1}$ denote the numbers

of supplier-side and buyer-side partners observed in the matched TDB-IIP sample in the previous year, and let P_{it}^S and P_{it}^B be the corresponding partners' patent applications in year t . Thus $P_{it}^S/S_{i,t-1}$ and $P_{it}^B/B_{i,t-1}$ measure patenting per pre-existing matched partner. Our estimating equation is

$$\log P_{it} = \beta_S \log \left(\frac{P_{it}^S}{S_{i,t-1}} \right) + \beta_B \log \left(\frac{P_{it}^B}{B_{i,t-1}} \right) + \alpha_i + \delta_t + \varepsilon_{it},$$

where α_i and δ_t are firm and year fixed effects. The log specification is estimated on firm-year observations with strictly positive own patent counts and strictly positive supplier-side and buyer-side partner-patenting measures.

To instrument partner patenting, we use foreign patenting to capture technological developments in the fields in which trading partners are active.¹ Using partners' lagged patent-class shares, we construct supplier- and buyer-side shift-share exposures as

$$Z_{it}^S = \sum_c \frac{P_{ic,t-1}^S}{P_{i,t-1}^S} \times P_{ct}^{\text{for}},$$

$$Z_{it}^B = \sum_c \frac{P_{ic,t-1}^B}{P_{i,t-1}^B} \times P_{ct}^{\text{for}}.$$

Here, P_{ct}^{for} counts triadic patent families in class c whose first JPO application occurs in year t and that have at least one applicant with a recorded country other than Japan. The IV regressions use the logarithms of these exposures, and the common estimation sample requires both exposures to be positive.

Table 1 reports ordinary least squares (OLS) and instrumental variable (IV) estimates. The coefficients on both supplier-side and buyer-side partner patenting are positive and precisely estimated. The OLS coefficients are around 0.02. In the joint IV specification, the supplier and buyer coefficients are 0.074 and 0.173, respectively. Because the specification is log-log, a 10 percent increase in supplier or buyer patenting is associated with approximately a 0.7 or 1.7 percent increase in the focal firm's own patenting. First-stage F-statistics range from 55.3 to 196.8. Taken together, these estimates suggest that firms patent more when their trading partners patent more.

The supplier and buyer coefficients remain positive and statistically significant when we exclude patents citing trading partners, control for the focal firm's own exposure to foreign

¹Acemoglu et al. (2016b) use foreign patenting to study the propagation of technological changes through production networks. For the econometric framework for shift-share instruments and shock-based identification, see Borusyak et al. (2022).

Table 1: Trading-Partner Patenting and Own Patenting

	OLS			IV		
	(1)	(2)	(3)	(4)	(5)	(6)
Supplier patenting per supplier	0.019*** (0.003)		0.017*** (0.003)	0.083*** (0.025)		0.074*** (0.026)
Buyer patenting per buyer		0.021*** (0.003)	0.019*** (0.003)		0.175*** (0.040)	0.173*** (0.041)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
First-stage F				196.8	108.0	55.3
Observations	50,918	50,918	50,918	50,918	50,918	50,918

Notes: The dependent variable is log patent applications. Supplier-side patenting is supplier patenting per matched supplier observed in the previous year’s TDB-IIP network; buyer-side patenting is buyer patenting per matched buyer observed in the previous year’s TDB-IIP network. Both are measured in logs. All columns use a common IV sample with strictly positive own patent counts and strictly positive supplier-side and buyer-side partner-patenting measures; the source panel covers 1996–2019. The excluded instruments in the IV columns are log shift-share exposures to foreign triadic patent families by patent class. First-stage F reports the weak-identification F-statistic from the clustered IV specification. Parentheses report standard errors clustered at the industry $\times$ year level. Significance levels are denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

patenting, or include industry-by-year fixed effects. Appendix Table A1 compares these specifications and the specification combining citation exclusions with the own-exposure control.

The positive relationships on both the supplier and buyer sides motivate examining the connection between trading partners’ technological activity and a firm’s innovation decisions. Guided also by the evidence on gradual link accumulation and relationship persistence (Appendix A.1), our model analyzes how supplier and buyer relationships shape the returns to innovation through input costs and demand.

3 A Model of Production Networks and R&D

We combine the product-line innovation framework of Klette and Kortum (2004) with endogenous firm entry and accumulating buyer–supplier relationships. Successful incumbent R&D creates new product lines that inherit the firm’s existing suppliers and buyers, making innovation returns depend on its accumulated network. Innovation and entry, in turn, shape the evolution of the production network.

3.1 Model Environment

A representative infinitely lived household supplies two distinct labor endowments: one unit of production labor and one unit of R&D labor. Production labor is used in goods production and earns wage $w(t)$, while R&D labor is used only for innovation and earns wage $w_H(t)$.²

Households have preferences over a final consumption good,

$$U_0 = \int_0^\infty \exp(-\rho t) \log Y(t) dt,$$

where $\rho > 0$ is the discount rate and $Y(t)$ is consumption. The household faces the budget constraint

$$\dot{A}(t) \leq r(t)A(t) + w(t) + w_H(t) + \Pi(t) - P^F(t)Y(t),$$

together with the standard no-Ponzi condition, where $A(t)$ is the asset position, $r(t)$ is the interest rate, $w(t)$ and $w_H(t)$ denote wages for production and R&D labor, respectively, $P^F(t)$ is the final-consumption price index, and $\Pi(t)$ denotes aggregate net payouts from firms owned by the household. Since these payouts are lump-sum from the household's perspective, they do not affect the Euler equation. We normalize nominal final expenditure to one at every date, $P^F(t)Y(t) = 1$. With log utility, the Euler equation implies that the interest rate in these units satisfies $r(t) = \rho$.

We characterize a stationary equilibrium and suppress time subscripts when there is no ambiguity.

Product Lines, Firms, and Production Networks

There is a continuum of differentiated product lines, indexed by $\omega \in \mathcal{P}(t)$. A product line is a differentiated variety; the measure of product lines evolves through creation and exit. Each product line ω is produced by a monopolist and can be used both as an intermediate input and as a final-consumption variety.

A firm may own and produce multiple product lines simultaneously. A firm's age is its time since entry. Consider a firm i that owns product line ω , and let $n(i)$ denote the number of product lines owned by firm i . We drop the firm subscript when no confusion arises.

Denote by $\mathcal{S}(\omega) \subset \mathcal{P}$ the set of product lines used as inputs for product line ω . Although

²The unit supply of R&D labor is a normalization. Measuring R&D labor in units of the aggregate endowment absorbs its scale into the R&D efficiency parameters. The fixed endowment makes the main question allocative, following recent work on R&D allocation and misallocation (e.g., Liu and Ma, 2026).

buyers can be inferred from the inverse mapping of $\mathcal{S}(\cdot)$, for notational simplicity we write $\mathcal{B}(\omega) \subset \mathcal{P}$ for the set of buyers of product line ω . Under the matching and network-inheritance structure introduced below, a product line will be summarized by the age of its owner firm rather than by its own time since birth.

The empirical network is observed at the firm-pair level, whereas the model tracks product-line relationships. The calibration maps product-line link exposure into firm-level network moments by aggregating product-line link masses and normalizing by firm mass.³

Production and Contracting Given Networks

Firms use production workers and intermediate inputs for production. Intermediate inputs are imperfect substitutes with a constant elasticity of substitution, $\sigma > 1$. Production workers and the composite of intermediate inputs are combined in a Cobb-Douglas aggregator with labor share $\beta \in (0, 1)$:

$$x(\omega) = \frac{1}{\beta^\beta (1-\beta)^{1-\beta}} l(\omega)^\beta \left(\int_{\omega' \in \mathcal{S}(\omega)} x(\omega', \omega)^{\frac{\sigma-1}{\sigma}} d\omega' \right)^{\frac{\sigma}{\sigma-1}(1-\beta)}, \quad (1)$$

where $l(\omega)$ is the demand for production workers used to produce product line ω , and $x(\omega', \omega)$ is the demand for product line ω' used to produce product line ω . The integral is taken only over linked suppliers in $\mathcal{S}(\omega)$, so a firm's network directly determines the set of intermediate varieties available for production.

A representative final-good producer aggregates product lines with the same CES elasticity of substitution σ as in the intermediate-input aggregator:

$$Y = \left(\int_{\omega \in \mathcal{P}} y(\omega)^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}},$$

where $y(\omega) = Y \left(\frac{p^F(\omega)}{P^F} \right)^{-\sigma}$ is the final demand for product line ω . Gross output of a product line is used both as an input by its buyers and as a final consumption variety. Thus, for all ω , $x(\omega)$ satisfies the following market clearing condition:

$$x(\omega) = \int_{\omega' \in \mathcal{B}(\omega)} x(\omega, \omega') d\omega' + y(\omega),$$

The main specification uses a two-part-tariff contracting environment on buyer-supplier links. The final-good producer represents household demand served through linear

³When multiple inherited product-line links correspond to the same firm-level trading partner, the model counts exposure at the product-line level. Appendix B.1 describes this measurement convention.

pricing, whereas established buyer–supplier relationships permit separate contracting over marginal input prices and fixed transfers. The maintained contracting assumptions are that marginal prices govern static quantities and that fixed transfers allocate the resulting bilateral operating surplus. Let $c(\omega)$ denote unit production cost. Intermediate inputs are priced at marginal cost $c(\omega)$, while final-consumption sales use

$$p^F(\omega) = \mu^F c(\omega).$$

Contracts are end-use contingent. Product-line producers purchase intermediate inputs at intermediate-input prices, while the final-good producer purchases varieties at final-consumption prices. Resale across uses is ruled out. The fixed-transfer rule assigns suppliers a share $\theta \in [0, 1]$ of bilateral operating surplus, as formalized in the static contracting problem below. Since these transfers are fixed with respect to contemporaneous input quantities, they affect private product-line payoffs but not the static input allocation or aggregate resource feasibility. We set $\mu^F = \sigma/(\sigma - 1)$ and $\theta = 1/2$.

Innovation, Entry, Network Inheritance, and Exit

Firms conduct R&D to acquire new product lines.⁴ Our central assumption is that each new product line reuses its parent product line’s trading relationships at creation.

Assumption 1. *At the creation of a new product line ω' derived from the incumbent firm’s product line ω ,*

$$\mathcal{S}(\omega') = \mathcal{S}(\omega),$$

$$\mathcal{B}(\omega') = \mathcal{B}(\omega).$$

Thus, a new product line inherits its parent line’s supplier and buyer links at creation. Subsequent matching occurs at the product-line level, and individual links dissolve at rate δ_M . Inheritance makes the value of incumbent R&D depend on the trading relationships accumulated by the parent product line.

Following Klette and Kortum (2004), a firm with n product lines that chooses per-product-line innovation rate λ incurs total R&D cost $nw_H h_H(\lambda)$, where

$$h_H(\lambda) = \frac{1}{\varphi} \lambda^\gamma.$$

⁴The product-line formulation follows Klette and Kortum (2004) and focuses on expansion into additional lines, which is the margin through which network inheritance affects incumbent R&D in the model.

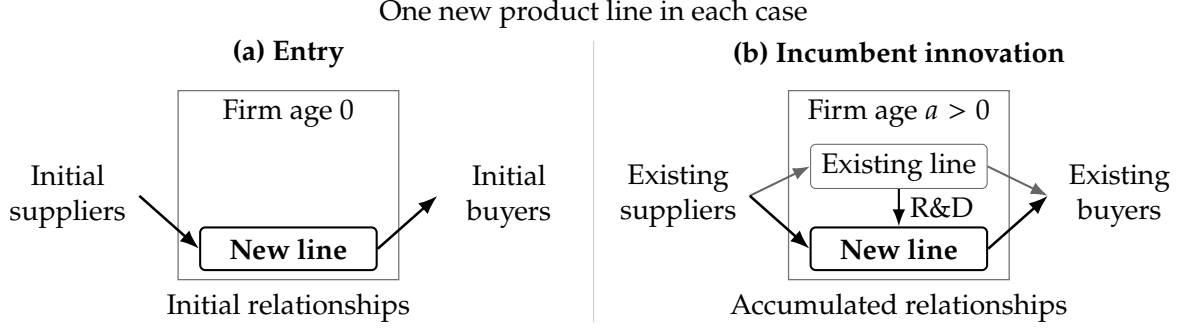


Figure 2: Network Inheritance in Entry and Incumbent Innovation

Notes: Both panels show the creation of a new product line. An entrant begins with initial supplier and buyer relationships, governed by ζ_0 . A new line created by an incumbent inherits the existing line's suppliers and buyers, while the existing line remains in operation. Firm outlines identify common ownership. Supply arrows run from suppliers to product lines and from product lines to buyers; the R&D arrow denotes product-line creation.

Here $\varphi > 0$ is the efficiency parameter and $\gamma > 1$.⁵ A firm with n product lines therefore creates new product lines at total rate $n\lambda$.

New firm creation uses the R&D labor resource. A flow E of new firms enters, with aggregate R&D labor requirement⁶

$$H_E(E) = \frac{E^{\gamma_E}}{\varphi_E}, \quad \gamma_E > 1, \varphi_E > 0.$$

Each entrant starts with one age-zero product line. The parameter ζ_0 controls the initial buyer-supplier link intensity at entry. We set $\gamma_E = \gamma = 2$, so entry and incumbent R&D have the same R&D-cost curvature but distinct efficiency parameters.

Figure 2 contrasts the networks available to new product lines created through entry and incumbent innovation. Both activities create a new variety, but incumbent innovation draws on relationships accumulated by the owner firm.

New buyer-supplier relationships form through random matching with efficiency ζ . Existing relationships are severed at rate $\delta_M > 0$. Firms exit exogenously at rate δ_F , removing all their product lines and attached trading links. Product lines exit exogenously at rate $\delta_P > 0$; product-line exit removes the line and its attached trading links but does not

⁵This product-line-level cost function can be micro-founded by a constant-returns-to-scale innovation technology that uses R&D workers and the number of product lines as inputs. Namely, a firm i hires $l_H(i)$ units of R&D workers to add one more product line at the flow rate $\Lambda(i) = n(i)^{1-1/\gamma} (\varphi l_H(i))^{1/\gamma}$, where Λ is the firm-level flow rate.

⁶This specification can be generated by a fixed pool of potential entry projects, each exhibiting diminishing returns to R&D. The size of the pool is absorbed into φ_E .

by itself change the mass of firms.⁷ After entry, firms face the same matching, innovation, product-line exit, and firm-exit processes as incumbents. Write $d = \delta_F + \delta_P$ for the total exit rate of a product line.

3.2 Stationary Decentralized Equilibrium

The stationary decentralized equilibrium jointly determines firm entry, incumbent R&D, and the age structure of production networks. Entry and incumbent R&D determine the product-line density over firm age. The matching technology maps this density into age-indexed buyer-supplier links. The static contracting block maps those links into unit costs, revenues, bilateral surplus, and private product-line payoffs. The dynamic problem then maps those payoffs back into incumbent R&D and entry choices.

Product-Line Distribution and Matching

In the model's maintained age-indexed matching structure, a product line faces deterministic masses of suppliers and buyers indexed by their owner-firm ages. Given the aggregate network state, product lines owned by firms of the same age face common partner-age distributions, without requiring identical partner sets. Newly created lines inherit these distributions from their parent lines. This age-indexed structure and network inheritance allow us to summarize a product line by the age of its owner firm, $a \geq 0$, rather than by the line's own time since birth. Firms of the same age can differ in size because their numbers of product lines evolve stochastically through innovation and product-line exit.

Let $f(a, t)$ denote the density of product lines owned by age- a firms at time t and $N(t) \equiv \int_0^\infty f(a, t) da$ the aggregate product-line stock. In the stationary equilibrium we drop time arguments and write $f(a)$ and N .

Entrants arrive at flow rate E , and each entrant starts with one age-zero product line. Let N_f denote the mass of surviving firms. In stationarity,

$$N_f = \frac{E}{\delta_F}. \quad (2)$$

Matching depends on the aggregate product-line stock rather than firm mass. Firm mass is used to report objects such as links per firm and to map model link masses into the empirical moments used in the calibration.

⁷For accounting, a firm that loses all product lines remains in the firm population as an inactive zero-product firm, with $V^F(0, a) = 0$ and no production or R&D activity until firm exit.

The stationary density of product lines owned by age- a firms satisfies

$$\frac{\partial f(a)}{\partial a} = (\lambda(a) - d) f(a), \quad a > 0, \quad (3)$$

together with the boundary condition

$$f(0) = E. \quad (4)$$

The term $\lambda(a)f(a)$ does not enter as an inflow at age zero because age indexes the owner firm rather than the product line itself: successful R&D by an age- a firm creates an additional product line owned by an age- a firm. The law of motion implies

$$f(a) = E \exp \left\{ \int_0^a (\lambda(s) - d) ds \right\}.$$

Thus E controls the scale of the product-line density, while $\lambda(a)$ controls the age profile.

Relationships are observed at the firm-pair level, whereas the search opportunities behind those relationships are product-line needs and capabilities. We maintain constant returns in aggregate matching. Let aggregate meetings be proportional to N . Dividing first by the mass of buyer product lines and then allocating meetings across supplier ages yields age-specific access $f(a_s)/N$. The coefficients ζ and ζ_0 absorb matching efficiency. A proportional increase in the product-line density therefore leaves per-line meeting rates unchanged.

Stationary Matched-Product Density

The next lemma is the main tractability result for the matching block. Once the distribution of product lines across firm ages is known, it gives link exposure by supplier and buyer ages without tracking each firm's individual partner history.

Lemma 1 (Stationary Matched-Product Density and Reciprocity). *Let $m(a_s, a_b)$ denote the supplier-line density per buyer product line, where a_s and a_b are the ages of the supplier and buyer firms. Equivalently, $m(a_s, a_b) da_s$ is the mass of supplier product lines whose owner firms have ages in $[a_s, a_s + da_s]$ and that are connected to one buyer product line owned by an age- a_b firm. Then $m(a_s, a_b)$ satisfies*

$$\underbrace{\frac{\partial}{\partial a_s} m(a_s, a_b) + \frac{\partial}{\partial a_b} m(a_s, a_b)}_{\text{time evolution of the matched-product density}}$$

$$= \underbrace{-(\delta_M + d) m(a_s, a_b)}_{\text{link and supplier-line destruction}} + \underbrace{\zeta \frac{f(a_s)}{N}}_{\text{random matching}} + \underbrace{\lambda(a_s) m(a_s, a_b)}_{\text{inherited links from supplier R\&D}}, \quad a_s, a_b > 0, \quad (5)$$

where $\lambda(a_s)$ is the innovation rate of an age- a_s supplier firm, with two entry boundaries:

$$\begin{aligned} m(a_s, 0) &= \zeta_0 \frac{f(a_s)}{N} \\ m(0, a_b) &= \zeta_0 \frac{f(0)}{N} = \zeta_0 \frac{E}{N}. \end{aligned}$$

The first boundary condition assigns suppliers to entrant buyers according to the incumbent product-line density. The second assigns entrant suppliers to incumbent buyer lines at a rate proportional to the entry flow. Given f , the stationary closed form is

$$m(a_s, a_b) = \left[\frac{\zeta}{\delta_M} - \left(\frac{\zeta}{\delta_M} - \zeta_0 \right) \exp\{-\delta_M \min\{a_s, a_b\}\} \right] \frac{f(a_s)}{N}. \quad (6)$$

As the younger firm ages, the bracketed term converges from the initial link intensity ζ_0 to the long-run level ζ/δ_M . Because the bracketed term in (6) depends on ages only through the younger endpoint, $\min\{a_s, a_b\}$, the same exposure can be read from the buyer or supplier side:

$$\frac{m(a_s, a_b) f(a_b)}{f(a_s)} = m(a_b, a_s),$$

For a given buyer age a_b , the interior law of motion has three flows. First, the mass of linked supplier product lines increases when an age- a_s supplier's R&D creates a new product line and the new product line inherits the supplier's existing buyer links. Second, linked supplier product lines are lost when the relationship is severed at rate δ_M , when the supplier firm exits at rate δ_F , or when the supplier product line exits at rate δ_P .⁸

Third, random matching adds age- a_s supplier product-line mass at rate $\zeta f(a_s)/N$. Matching weights supplier ages by the product-line mass $f(a_s)$.

Entry appears through the two boundary conditions. New buyers at $a_b = 0$ draw $\zeta_0 f(a_s)/N$ supplier product-line mass from age- a_s suppliers. Similarly, entrant suppliers have one product line and are connected to existing buyer product lines at rate $\zeta_0 E/N$.

The reciprocity relation converts supplier access per buyer line into buyer access per supplier line. The normalization is per buyer product line throughout: multiplying $m(a_s, a_b)$ by the mass of buyer product lines of age a_b would recover the total mass of links

⁸Buyer-side firm or product-line exit removes the buyer line itself and is handled by the product-line density $f(a_b)$; it is not an additional loss term in $m(a_s, a_b)$, which is measured per buyer product line.

from age- a_s suppliers to age- a_b buyers. Accordingly, when an age- a_s line is viewed as a supplier, the buyer mass of age a_b attached to it is written as $m(a_b, a_s)$.

Static Contracting Problem

Taking the product-line density and matched-product density (f, m) as given, the decentralized static contracting problem maps the age-indexed network into the private payoff $\pi(a)$ that enters the dynamic R&D problem. It has two parts. The first part determines marginal prices, input shares, unit costs, revenues, and variable production costs per product line $C(a)$, normalized by final expenditure. The second part applies the fixed-transfer rule and converts bilateral operating surplus into the private flow payoff. Fixed transfers affect private payoffs but do not change static input choices. We suppress the environment superscript DE in this subsection and restore it when comparing allocations.

Prices and quantities. Intermediate inputs are priced at unit production cost $c(a)$, while the final-consumption price is

$$p^F(a) = \mu^F c(a).$$

Unit costs satisfy

$$c(a_b) = w^\beta \left(\int c(a_s)^{1-\sigma} m(a_s, a_b) da_s \right)^{\frac{1-\beta}{1-\sigma}}. \quad (7)$$

The final-consumption price index and final-sales revenue, normalized by final expenditure, are

$$P^F = \left(\int p^F(a)^{1-\sigma} f(a) da \right)^{\frac{1}{1-\sigma}},$$

$$R^F(a) = \left(\frac{p^F(a)}{P^F} \right)^{1-\sigma}.$$

For a supplier of age a_s and a buyer of age a_b , the input-expenditure weight per supplier variety is

$$s^M(a_s, a_b) = \frac{c(a_s)^{1-\sigma}}{\int c(\tilde{a}_s)^{1-\sigma} m(\tilde{a}_s, a_b) d\tilde{a}_s}.$$

These weights satisfy $\int s^M(a_s, a_b) m(a_s, a_b) da_s = 1$. The input-expenditure share of suppliers in the age interval $[a_s, a_s + da_s]$ is $s^M(a_s, a_b) m(a_s, a_b) da_s$. Let $x(a)$ denote gross

output per product line. Its variable production cost, normalized by final expenditure, is

$$C(a) \equiv \frac{c(a)x(a)}{P^F Y}.$$

By the reciprocity relation above, the mass of age- a_b buyers attached to an age- a_s supplier line is $m(a_b, a_s)$. Hence intermediate-sales revenue per age- a_s product line is

$$R^M(a_s) = (1 - \beta) \int C(a_b) s^M(a_s, a_b) m(a_b, a_s) da_b. \quad (8)$$

Dividing final-sales revenue by its markup recovers the associated variable production cost. Intermediate-sales revenue equals the cost of supplying those inputs. Market clearing therefore gives the recursion for variable production costs and the production-labor clearing condition:

$$C(a_s) = \frac{R^F(a_s)}{\mu^F} + R^M(a_s), \quad (9)$$

$$w = \beta \int C(a) f(a) da. \quad (10)$$

Aggregating the variable-cost recursion under the normalization $P^F Y = 1$ gives

$$\int C(a) f(a) da = \frac{1}{\mu^F} + (1 - \beta) \int C(a) f(a) da.$$

Production-labor market clearing therefore implies $w = 1/\mu^F$. Given (f, m) , we first solve the unit-cost fixed point at this wage. Costs determine input shares and final-sales revenues, after which the linear recursion determines $C(a)$. Bilateral transfers and private flow payoffs then follow.

Transfers and private payoffs. The marginal prices above determine quantities, costs, and revenues. The remaining part of the assumed two-part tariff is the fixed transfer. Given the static allocation, the bilateral operating surplus generated by an age- a_s supplier linked to an age- a_b buyer is

$$S(a_s, a_b) = \frac{1 - \beta}{\sigma - 1} C(a_b) s^M(a_s, a_b), \quad (11)$$

This surplus measures the buyer's marginal variable-cost saving from access to an additional supplier variety, holding its output and input prices fixed. Under CES input demand, it equals expenditure on that variety divided by $\sigma - 1$. Under marginal-cost pricing, the

supplier earns no variable profit on the transaction, so this cost saving measures the bilateral operating surplus before fixed transfers.

The contracting assumption is that the flow transfer paid by buyer a_b to supplier a_s equals $\theta S(a_s, a_b)$. Thus, fixed transfers reallocate current bilateral operating surplus between linked buyers and suppliers without changing the static input choices characterized above.

Intermediate inputs are priced efficiently at the margin, while final-consumption sales still carry the monopolistic markup $\mu^F = \sigma/(\sigma - 1)$.

The private flow payoff generated by a product line owned by an age- a firm is

$$\pi(a) = \underbrace{\frac{R^F(a)}{\sigma}}_{\text{Final-sales profit}} + \underbrace{\theta \int S(a, a_b) m(a_b, a) da_b}_{\text{Transfers from buyers}} - \underbrace{\theta \int S(a_s, a) m(a_s, a) da_s}_{\text{Transfers to suppliers}}. \quad (12)$$

Firm Value, Innovation, and Entry

Since the product-line flow payoff depends on owner-firm age through the inherited network, the firm's optimization problem can also be summarized by age and the number of product lines. A firm with $n \geq 1$ product lines and age a maximizes the value $V^F(n, a)$, with $V^F(0, a) = 0$:

$$\begin{aligned} (\rho + \delta_F) V^F(n, a) = & \underbrace{n\pi(a)}_{\text{flow payoff}} + \underbrace{V_a^F(n, a)}_{\text{age effect}} \\ & + \underbrace{n\delta_P \{V^F(n-1, a) - V^F(n, a)\}}_{\text{product-line exit}} \\ & + \max_{\lambda \geq 0} \left[\underbrace{n\lambda \{V^F(n+1, a) - V^F(n, a)\}}_{\text{product-line expansion}} - \underbrace{nw_H h_H(\lambda)}_{\text{R\&D costs}} \right]. \end{aligned} \quad (13)$$

The term $\delta_F V^F(n, a)$ reflects exogenous firm exit, which destroys all product lines owned by the firm. On the right-hand side, the first term is total private flow payoff, the second term is the change in firm value due to aging, the third term is the value loss from product-line exit at total rate $n\delta_P$, and the final bracket is the net gain from adding new product lines through R&D.

Linearity in the number of product lines implies that firm value can be expressed as the sum of product-line values. Flow payoffs, product-line exit risk, and R&D costs all scale with the number of product lines. The maintained age-indexed matching structure

and network inheritance give all product lines within an age- a firm common partner-age distributions and the same per-line payoff $\pi(a)$. We therefore conjecture

$$V^F(n, a) = nV(a)$$

where $V(a)$ is the value of a product line owned by an age- a firm. Substituting into (13) gives the HJB equation for $V(a)$:

$$(\rho + d) V(a) = \pi(a) + V_a(a) + \max_{\lambda \geq 0} [\lambda V(a) - w_H h_H(\lambda)] \quad (14)$$

The left-hand side captures discounting, owner-firm exit, and product-line exit. On the right-hand side, the first term is the static private payoff from a product line, the second term is the effect of aging on product value, and the maximization term is the net value of innovation, where $\lambda V(a)$ is the value of product-line expansion and $w_H h_H(\lambda)$ is the associated R&D cost. The first-order condition yields an optimal innovation rate:

$$\lambda(a) = \left\{ \frac{\varphi}{\gamma w_H} V(a) \right\}^{\frac{1}{\gamma-1}}, \quad (15)$$

The innovation rate depends on the value of the entire inherited supplier and buyer portfolio. Changes in the value of these relationships affect a common R&D choice and hence the rate at which the portfolio is replicated through product-line creation. The innovation problem therefore cannot be decomposed into independent link-level decisions.⁹ Our age-indexed matching structure and network inheritance make this joint problem tractable, allowing the inherited portfolio to be summarized by owner-firm age conditional on the aggregate network state.

The production labor market clears inside the static contracting block through (10). Since $h_H(\lambda(a))$ is incumbent R&D labor used per product line, the aggregate R&D labor market clearing condition is

$$1 = H_E(E) + \int h_H(\lambda(a)) f(a) da = \frac{E^{\gamma_E}}{\varphi_E} + \int \frac{\lambda(a)^{\gamma}}{\varphi} f(a) da. \quad (16)$$

Free entry equates the private value of an entrant's initial product line to the marginal

⁹In Huneus (2020), sellers' revenues are additive across buyers and sellers pay link-specific fixed costs, allowing relationships to be evaluated separately conditional on the equilibrium path of bilateral profits. Here, R&D is chosen for the inherited portfolio as a whole.

R&D labor cost of entry:

$$V(0) = w_H H'_E(E) = w_H \frac{\gamma_E}{\varphi_E} E^{\gamma_E-1}. \quad (17)$$

The corresponding Kuhn–Tucker form is $E \geq 0$, $w_H H'_E(E) \geq V(0)$, and $E[w_H H'_E(E) - V(0)] = 0$. Equivalently, when the entry margin is interior,

$$E = \left\{ \frac{\varphi_E}{\gamma_E w_H} V(0) \right\}^{\frac{1}{\gamma_E-1}}.$$

Equilibrium Definition

A stationary decentralized equilibrium is a collection of functions and scalars

$$\{f, m, c, C, V, \lambda, E, w, w_H\}$$

such that:

1. Given the private flow payoff in (12), the value function V and innovation policy λ satisfy (14) and (15).
2. Entry satisfies (17).
3. Given (λ, E) , the product-line density f and matched-product density m satisfy (3), (4), and the matching law of motion (5) with entrant boundary conditions, equivalently the closed form (6).
4. Given (f, m) , the static contracting block determines the derived objects $p^F, P^F, R^F, R^M, s^M, S$, and π through (7)–(12).
5. R&D labor clears according to (16), with production labor clearing embedded in (10).

These equilibrium objects form the baseline for the comparison of private and social innovation values in the next section.

4 R&D Misallocation

Having established the decentralized equilibrium in Section 3, we compare the private value that guides decentralized R&D with the social value assigned by the planner. These values determine aggregate entry and the incumbent R&D profile across firm ages. The

planner values product lines together with their inherited relationships and internalizes how product-line creation changes matching opportunities throughout the network. To simplify notation, we suppress environment superscripts on endogenous variables such as (f, m, c, C) when no confusion arises.

4.1 Product-Line Values and R&D Allocation

This subsection translates product-line and entrant values into the two R&D allocation margins used below: aggregate entry and the age profile of incumbent R&D. The decentralized equilibrium and the planner allocations share the same allocation structure once the relevant values are specified. Let j index these allocations, with $V^{DE} \equiv V$ and $w_H^{DE} \equiv w_H$. In each allocation, the entrant value is $V^j(0)$, including the initial trading relationships through the entry boundary conditions.

The common allocation rule follows directly from the R&D cost functions and the fixed R&D resource constraint. In the decentralized equilibrium and in each planner allocation satisfying the corresponding entry and incumbent R&D optimality conditions, entry and incumbent R&D compete for the fixed R&D resource. Entry satisfies the Kuhn–Tucker condition

$$E^j \geq 0, \quad w_H^j H'_E(E^j) \geq V^j(0), \quad E^j \left[w_H^j H'_E(E^j) - V^j(0) \right] = 0. \quad (18)$$

The residual amount of R&D labor available to incumbents is

$$H_H^j \equiv 1 - H_E(E^j).$$

Conditional on this residual labor, if incumbent R&D is interior and $h_H(\lambda) = \lambda^\gamma / \varphi$, per-product-line incumbent R&D labor for product lines owned by age- a firms satisfies

$$h_H \left(\lambda^j(a) \right) = H_H^j \frac{(V^j(a))^{\frac{\gamma}{\gamma-1}}}{\int (V^j(\tilde{a}))^{\frac{\gamma}{\gamma-1}} f^j(\tilde{a}) d\tilde{a}}. \quad (19)$$

This characterization separates the entry margin from the allocation of R&D across incumbent ages. Conditional on the labor devoted to entry, the remaining R&D labor is allocated among incumbent product lines in proportion to $[V^j(a)]^{\gamma/(\gamma-1)}$. The total labor allocated to an age group also reflects the number of product lines in that group. In the decentralized equilibrium, $V^j = V$ is the private product-line value defined in (14), and the entry condition is the free-entry condition in (17). The planner instead uses the social

product-line values derived below.

4.2 Social Planner Values

The first-best social planner maximizes welfare by internalizing all production, entry, and matching-stock effects in the economy. Unlike the decentralized equilibrium, the social planner directly controls goods production, incumbent R&D, and entry.

The social planner maximizes the discounted utility flow

$$U_0 = \int_0^\infty \exp(-\rho t) \log Y(t) dt$$

where $Y = \left(\int y(a)^{\frac{\sigma-1}{\sigma}} f(a) da \right)^{\frac{\sigma}{\sigma-1}}$. The planner chooses goods production, incumbent R&D, and entry subject to the age-notation production technology (20), goods-market clearing (21), the production labor resource constraint (22), the laws of motion (5) and (3), the entry boundary condition (4), the entrant boundary conditions for m , and the R&D resource constraint (16). Because the intratemporal allocation is static for given (f, m) , the problem separates into a static goods-allocation block and a dynamic R&D-and-entry block. The full derivation is in Section F.

For a fixed network state (f, m) , the planner evaluates production on the same buyer-supplier network as the decentralized economy. Production is valued at social marginal cost, and fixed transfers do not enter resource feasibility. In the baseline, intermediate-input marginal prices are efficient and the production wage adjusts to the uniform final-consumption markup, so the decentralized and planner static goods allocations coincide at a given (f, m) . This static efficiency is consistent with a gap between private profits and social innovation returns. The welfare gains from moving to the planner allocation therefore arise through changes in R&D allocation and the resulting evolution of product lines and trading relationships. The static problem below identifies the social cost and demand objects that determine the planner current-return term $R^{SP}(a)$ in the dynamic R&D problem.

The planner's intratemporal constraints use a_s for supplier age and a_b for buyer age:

$$x(a_b) = \frac{1}{\beta^\beta (1-\beta)^{1-\beta}} l(a_b)^\beta \left(\int x(a_s, a_b)^{\frac{\sigma-1}{\sigma}} m(a_s, a_b) da_s \right)^{\frac{\sigma}{\sigma-1}(1-\beta)}, \quad (20)$$

$$x(a_s) f(a_s) = \int x(a_s, a_b) m(a_s, a_b) f(a_b) da_b + y(a_s) f(a_s), \quad (21)$$

$$\int l(a) f(a) da = 1 \quad (22)$$

Here $x(a)$ is gross output of a product line owned by an age- a firm, $y(a)$ is final-good use, $l(a)$ is production labor, and $x(a_s, a_b)$ is the quantity supplied by an age- a_s supplier product line to an age- a_b buyer product line.

The associated static objects can be summarized by a social unit cost c^{SP} and a demand index D^{SP} :

$$c^{SP}(a_b) = \left(\int c^{SP}(a_s)^{1-\sigma} m(a_s, a_b) da_s \right)^{\frac{1-\beta}{1-\sigma}},$$

$$D^{SP}(a_s) = (1 - \beta) \int [c^{SP}(a_b)]^{\frac{\beta}{1-\beta}(\sigma-1)} D^{SP}(a_b) \frac{m(a_s, a_b) f(a_b)}{f(a_s)} da_b + 1.$$

The planner price index is

$$P^{SP} = \left(\int c^{SP}(a)^{1-\sigma} f(a) da \right)^{\frac{1}{1-\sigma}}.$$

The social current return per product line is

$$R^{SP}(a) \equiv c^{SP}(a)x(a) = \left(\frac{c^{SP}(a)}{P^{SP}} \right)^{1-\sigma} D^{SP}(a).$$

This term combines the final-consumption contribution of a new product with the net input-variety benefits generated through its inherited buyer relationships. The input-expenditure weight per supplier variety is

$$s(a_s, a_b) \equiv \frac{c^{SP}(a_s)^{1-\sigma}}{\int c^{SP}(\tilde{a})^{1-\sigma} m(\tilde{a}, a_b) d\tilde{a}}.$$

Proposition 1 (Planner Product-Line and Match Values). *At a stationary planner allocation, let $V^{SP}(a)$ denote the marginal social value of a product line together with its inherited trading relationships.¹⁰ It satisfies*

$$\left(\rho + d - \lambda^{SP}(a) \right) V^{SP}(a) = \underbrace{R^{SP}(a)}_{\text{Current social return}} + \underbrace{\partial_a V^{SP}(a)}_{\text{Aging}}$$

¹⁰Social values and the shadow R&D wage are expressed in units that multiply their utility-based counterparts by $\sigma - 1$. Thus the current marginal benefit $R^{SP}(a)/(\sigma - 1)$ enters the normalized value equation as $R^{SP}(a)$; the derivation is in Appendix F.8.

$$\underbrace{-w_H^{SP} h_H(\lambda^{SP}(a))}_{\text{R\&D cost}} - \underbrace{\frac{\Omega}{N}}_{\text{Matching congestion}}. \quad (23)$$

Here w_H^{SP} is the shadow wage on R&D labor, and $-\frac{\Omega}{N}$ is the planner's aggregate matching-stock term, defined in Section 4.3.

The planner entrant value is $V^{SP}(0)$, with the entrant's initial buyer-supplier links incorporated through the entry boundary conditions.

Let $V^M(a_s, a_b)$ denote the social value, per product line owned by an age- a_b buyer firm, of adding a variety supplied by an age- a_s firm to that buyer's input menu. It satisfies

$$\begin{aligned} \left(\rho + 2d + \delta_M - \lambda^{SP}(a_s) - \lambda^{SP}(a_b) \right) V^M(a_s, a_b) = & V_{a_s}^M(a_s, a_b) + V_{a_b}^M(a_s, a_b) \\ & + (1 - \beta) R^{SP}(a_b) s(a_s, a_b), \end{aligned} \quad (24)$$

This match value is used to evaluate the aggregate matching-stock term $-\frac{\Omega}{N}$. Appendix F derives these value equations and the entrant value.

4.3 Private and Social Innovation Returns

The planner values enter the entry and incumbent-R&D allocation rules in (18) and (19). The gaps between private and social innovation returns arise from surplus appropriation and matching-stock congestion.¹¹

The first source concerns surplus appropriation. Private firms capitalize the future profits earned through inherited and newly formed relationships. A new supplier variety also expands buyers' input possibilities, while the supplier's contracts determine how much of the resulting surplus enters its own payoff. Final-consumption markups and bilateral surplus sharing determine private flow payoffs even when intermediate-input marginal prices are efficient. As Section F.5 shows, the social flow value of a match equals the buyer's benefit from input variety net of the resource cost of supplying the additional input.

The age pattern of this gap follows from the contracting rules. At a common network state, the ratio of private flow payoff to the planner's current return is

$$\frac{\pi(a)}{R^{SP}(a)} = \frac{1}{\sigma} \left[\theta\beta + \frac{1 - \theta}{D^{SP}(a)} \right]. \quad (25)$$

¹¹Mankiw and Whinston (1986) study entry biases arising from business stealing and product diversity. Here, accumulated supplier and buyer relationships shape the relative social returns to entry and incumbent product-line expansion.

With life-cycle network accumulation, buyer access and the demand index $D^{SP}(a)$ rise with firm age. The demand index reflects a firm's access to buyers and the demand they face through their own downstream relationships. As new product lines inherit these relationships, older firms' accumulated networks give their innovations a wider economic reach. For $\theta < 1$, the payoff ratio therefore declines with age. The decline reflects the growing importance of intermediate-input use relative to final-consumption sales in the social current return.¹² Private payoffs thus place less relative weight on older firms' social current returns than on those of entrants and young firms. Section F.9 derives this age ordering.

Full surplus appropriation ($\theta = 1$) makes this ratio equal to β/σ at every age.

The second source is matching-stock congestion. A new product line, whether created by entry or incumbent expansion, increases the denominator N in the matching rate. For a given mass of potential counterpart lines, this lowers the density of new matches directed toward those lines. Individual entrants and incumbents take the aggregate product-line stock as given. An innovator values the relationships its own products obtain, whereas the planner also values the changes in matching opportunities for other products. This congestion effect resembles business stealing, as new product lines divert future matching opportunities away from existing products.

The planner evaluates these changes using the match value V^M . The resulting matching-stock term is $-\Omega/N$, where Ω sums new-match flows weighted by their social values.

$$\begin{aligned} \Omega \equiv & \underbrace{\frac{\zeta}{N} \iint V^M(a_s, a_b) f(a_s) f(a_b) da_s da_b}_{\text{Between incumbents}} \\ & + \underbrace{\frac{\zeta_0 E}{N} \int V^M(a_s, 0) f(a_s) da_s}_{\text{Entrants as buyers}} + \underbrace{\frac{\zeta_0 E}{N} \int V^M(0, a_b) f(a_b) da_b}_{\text{Entrants as suppliers}}. \end{aligned} \quad (26)$$

When match values are positive, $\Omega > 0$, so the matching-stock term is negative.

Every product line adds one unit to N , so this current congestion cost is common across ages. Accumulated supplier and buyer relationships instead raise $R^{SP}(a)$, making the congestion cost smaller relative to the current return of a well-connected product line.

Correcting surplus appropriation removes the relative underweighting of returns from established buyer relationships. Internalizing matching-stock congestion requires weighing the common cost of product-line creation against its network-dependent benefits. Table 2

¹²In a symmetric CES benchmark with only final-consumer links, private profits and social current returns rise proportionally with consumer access.

Table 2: Sources of Private–Social Innovation–Return Gaps

Source	Economic mechanism	Model objects
Surplus appropriation	Markups and contracts determine how much of the surplus generated by product lines and trading relationships firms capture.	Private flow payoff $\pi(a)$ and planner current return $R^{SP}(a)$
Matching-stock congestion	Product-line creation dilutes other firms' matching opportunities.	Planner matching-stock term $-\Omega/N$

summarizes these two sources.

Proposition 2 (A welfare-improving R&D reallocation). *Consider an interior stationary decentralized equilibrium with accumulating trading relationships, $\zeta > \delta_M \zeta_0$, satisfying the regularity conditions in Appendix F.10. The equilibrium admits arbitrarily small, feasible temporary reallocations of R&D labor that raise discounted household welfare in each of the following directions.*

1. *From entry to incumbent product-line expansion.*
2. *From younger to older incumbent firms, holding the R&D labor allocated to entry unchanged.*

In the decentralized equilibrium, the marginal private return to R&D labor is equalized across entry and incumbent firms of all ages. The marginal social return evaluated at the same equilibrium, including subsequent network accumulation and the adjustment of product lines and trading relationships, instead rises with firm age. Small transfers of R&D labor from entry to incumbent expansion and from younger to older incumbents therefore improve welfare.

When network exposure is independent of firm age, the stationary decentralized and planner R&D allocations coincide (Appendix F.9). The allocation distortion thus reflects age differences in accumulated trading relationships rather than the presence of trading relationships alone. Section 5 quantifies the optimal reallocation and its welfare gains.

5 Estimation and Welfare Analysis

This section quantifies the welfare consequences of R&D misallocation through production networks. We first estimate and validate the model using Japanese firm-level data. We then compare decentralized and planner allocations to assess how surplus appropriation and matching congestion affect R&D allocation between entrants and incumbents and

across incumbent ages, and measure the resulting transition welfare gains. Finally, we evaluate how much of these gains a uniform subsidy to incumbent R&D can achieve.

5.1 Estimation and Model Validation

This subsection describes how we estimate the baseline decentralized economy and validate its life-cycle implications. We first summarize its fixed and externally calibrated parameters, estimate the remaining parameters by minimum distance, report the targeted fit, and then compare untargeted life-cycle profiles with the empirical patterns.

5.1.1 Fixed and Externally Calibrated Parameters

Before estimating the scale-normalized parameters below, we fix the static production and contracting parameters, dynamic exit and link-destruction rates, and R&D cost curvatures. These fixed values are reported in the lower panel of Table 3.

The CES elasticity is set to $\sigma = 3$, following Miyauchi (2024), and implies a final-consumption markup $\mu^F = 1.5$. Intermediate inputs are priced at cost, and current bilateral operating surplus is split equally, $\theta = 1/2$. The annual discount rate is set to $\rho = 0.05$. We set $\beta = 0.33$, guided by manufacturing’s value-added share in the 2015 Japanese input-output table (Ministry of Internal Affairs and Communications, 2019).

The dynamic turnover parameters are calibrated separately. The link destruction rate is set to $\delta_M = 0.07266$, based on the two- and three-year link-stock survival calibration discussed in Appendix B.4. We set the firm exit hazard to $\delta_F = 0.04$, using the average annual closure rate of establishments covered by employment insurance in Japan over fiscal years 1998–2019 as a proxy for firm exit (Ministry of Health, Labour and Welfare, various years).¹³ Product-line exit is set to $\delta_P = 0.06$ to match a benchmark product’s drop probability conditional on firm survival.¹⁴

We use quadratic R&D costs for both incumbent innovation and entry, setting $\gamma = \gamma_E = 2$. For incumbent innovation, this corresponds to a research elasticity of 0.5, as in Acemoglu et al. (2016a).

¹³The average annual closure rate is approximately 4.04%. Converting this probability into a constant continuous-time hazard gives approximately 0.0413, which we round to 0.04.

¹⁴In Table 8, column 1 of the January 2009 working-paper version of Bernard et al. (2009), the regression without fixed effects predicts a five-year product-drop probability of 0.256 when log relative product size and log relative product tenure are both zero. The sample consists of surviving multiple-product firms. Mapping this benchmark prediction into a constant continuous-time hazard gives $-\log(1 - 0.256)/5 \approx 0.0591$, which we round to 0.06.

5.1.2 Target Moments and Minimum-Distance Estimation

We estimate the matching rate ζ , initial connectivity at entry ζ_0 , and incumbent R&D efficiency φ using three moment blocks. Buyer-supplier links per firm primarily discipline matching efficiency, while supplier and buyer life-cycle coefficients discipline entrant connectivity relative to subsequent link accumulation. The average partner-patenting coefficient in Table 1 disciplines normalized incumbent R&D efficiency.

The target for average links per firm is 27.3, based on the supplier and buyer counts reported by Imani and Ohyama (2022). We estimate supplier and buyer age profiles from TDB C2TRD transaction records linked to firm characteristics for manufacturing firms in 1998–2019. The coefficients measure differences in $\log(1 + \text{mean degree})$ between firms at establishment ages 1–10 and firms aged 50 and above, controlling for year, prefecture, and industry. We average the supplier and buyer coefficients at each age to obtain ten life-cycle targets. They capture how young firms’ networks differ in size from those of mature firms. The regression target is the arithmetic mean of the supplier and buyer joint IV coefficients, approximately 0.124. Appendix B details the construction of these targets and their model counterparts.

We fix φ_E during estimation and report φ and φ_E in units such that decentralized steady-state entry equals one. Appendix B gives the conversion.

For each candidate value of φ , we estimate ζ and ζ_0 by minimizing

$$Q_N = \left[\log \left(\frac{L^{\text{model}}}{27.3} \right) \right]^2 + \frac{1}{10} \sum_{a=1}^{10} \left(\beta_a^{\text{model}} - \bar{\beta}_a^{\text{data}} \right)^2.$$

Here L^{model} is average link exposure per firm, $\bar{\beta}_a^{\text{data}}$ is the arithmetic mean of the supplier and buyer coefficients at age a , and β_a^{model} is their common model counterpart. The criterion combines the squared log deviation in average links with the mean squared deviation across the ten averaged life-cycle coefficients, giving the life-cycle block a combined weight of one. Among these fitted economies, we select the value of φ that matches the average partner-patenting coefficient, along with its associated estimates of ζ and ζ_0 .

The model counterpart to the partner-patenting target is the arithmetic mean of the supplier and buyer coefficients from an auxiliary OLS regression with firm and year effects. It regresses log own innovation intensity per product line, λ , on the logs of mean supplier and buyer innovation intensities in a simulated firm panel. The panel follows the model’s innovation, exit, matching, link-destruction, and inheritance events. We evaluate the fit in independent simulations not used in estimation. Appendix B.3 gives the simulation design.

Table 3: Estimated and Calibrated Parameters

Parameter	Symbol	Value
Estimated Parameters		
Matching rate	ζ	2.319
Initial connectivity at entry	ζ_0	11.62
Incumbent R&D efficiency	φ	0.1841
Entry R&D efficiency	φ_E	3.894
Fixed and Externally Calibrated Parameters		
CES elasticity	σ	3
Primary-factor share	β	0.33
Discount rate	ρ	0.05
Link destruction rate	δ_M	0.07266
Firm exit rate	δ_F	0.04
Product-line exit rate	δ_P	0.06
R&D cost curvature	γ	2
Entry cost curvature	γ_E	2
Final-consumption markup	μ^F	1.5
Supplier share of bilateral surplus	θ	0.5

Notes: The estimator uses the three moment blocks in Table 4. R&D efficiencies are reported in units such that decentralized steady-state entry equals one. Rates are annual continuous-time hazards.

5.1.3 Estimated Parameters and Model Fit

Table 3 reports the estimated and calibrated parameters. Initial connectivity at entry is 36.4% of the long-run connectivity benchmark ζ/δ_M . Entrants therefore begin with fewer connections per product line than older firms, reflecting the accumulation of trading relationships over the firm life cycle. The product-line-weighted average incumbent innovation hazard is approximately 0.067 per year.

Table 4 reports the targeted fit. The model matches the link level and the average partner-patenting coefficient closely. Across the ten averaged life-cycle coefficients, the root mean squared difference is 0.092 log points. Figure 3(a) compares the average profile with the model and also shows the separate supplier and buyer coefficients. In ten independent simulations not used in estimation, the average partner-patenting coefficient is 0.118, compared with the empirical target of 0.124.

5.1.4 Life-Cycle Fit and Untargeted Comparisons

Figure 3 compares the model with the network-age targets used in estimation and with trading-partner ages, which are not targeted.

Table 4: Targeted Moments and Model Fit

Moment	Data	Model
Buyer-supplier links per firm	27.3	27.3
Average partner-patenting coefficient	0.124	0.118
Average-profile RMSE, ages 1–10	—	0.092

Notes: RMSE is the root mean squared difference between the ten age-specific supplier-buyer average coefficients and their model counterparts. The model regression uses log innovation intensity per product line, $\log \lambda$, as the counterpart of log patent applications. The reported coefficient averages ten independent simulations not used in estimation.

Panel A shows a narrowing network gap relative to mature firms as young firms age. From age 1 to age 10, the average data coefficient rises by 0.684 log points, compared with 0.419 in the model. The empirical supplier coefficients are more negative than the buyer coefficients, a directional difference the model does not capture.

Panel B compares average supplier and buyer ages in the data and the model. From the 0–4 group to the 45–49 group, supplier age rises from 41.8 to 43.5 years and buyer age from 38.1 to 40.2 years, compared with 35.2 and 40.3 years in the model. The model understates partner ages among young firms and generates a steeper rise. Although partner ages are not targeted in estimation, the model captures their approximate average level and their broad upward pattern over the firm life cycle.

We also examine aggregate R&D intensity, which is not targeted in estimation. In the decentralized steady state, incumbent R&D expenditure amounts to 4.43% of aggregate product sales, compared with 4.28% for R&D-performing manufacturing enterprises in Japan in fiscal year 2019 (Statistics Bureau of Japan, 2020, Table 2-3).¹⁵

5.2 Planner Allocation and Welfare Decomposition

This subsection compares the decentralized equilibrium at the estimate in Table 3 with the social planner allocation. We report stationary entry and incumbent R&D and the welfare gain along the transition from the decentralized steady state. We then examine how the two innovation-return gaps shape R&D allocation and decompose the transition welfare gain by allocation margin.

¹⁵R&D intensity is measured as the ratio of total R&D expenditure to total sales. Model sales include both final-consumption and intermediate-good sales. Including entry R&D raises the model ratio to 6.01%. In the TDB–ZAIMUT data, the aggregate ratio among manufacturing firm-years with positive R&D expenditure over 1998–2019 is 1.94%. The official survey and TDB comparisons differ in sample coverage and period.

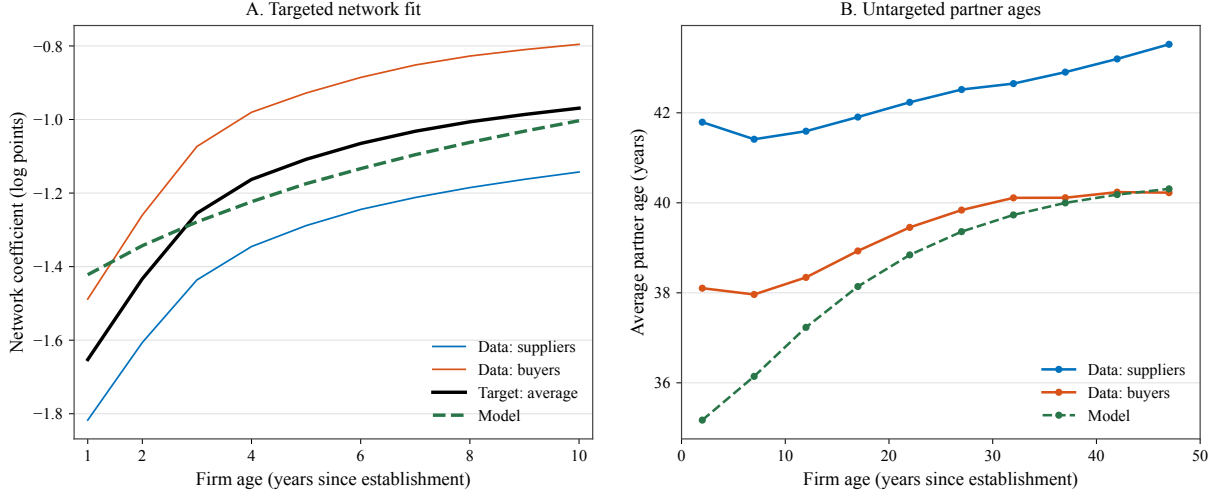


Figure 3: Model Fit to Network Life-Cycle Patterns

Notes: Panel A compares age coefficients for $\log(1 + \text{cell mean degree})$ at ages 1–10 relative to ages 50 and above. The thick solid line is the supplier-buyer average used in estimation; thin lines show the separate data series. Appendix B.2 defines the targets and their model counterparts. Panel B reports partner-age levels estimated jointly with five-year firm-age indicators, both endpoints' prefecture and industry effects, and year effects, weighting observed relationships equally. Age groups are shown at their midpoints. The model is symmetric across partner types, with partner ages weighted by link masses. Ages are measured from establishment using the TDB record year; both panels use 1998–2019 data.

5.2.1 Planner Allocation and Transition Welfare

Transition welfare gains are reported as consumption-equivalent (CE) changes. For a computed transition path, let U_{path} denote discounted utility along the reform path and U_{DE} discounted utility from remaining in the decentralized steady state. We define

$$\log(1 + \Delta_{\text{CE}}) = \rho (U_{\text{path}} - U_{DE}) . \quad (27)$$

Thus a reported transition CE gain is the permanent proportional increase in decentralized steady-state consumption that would make the household indifferent to the planner transition.

Relative to the planner, the decentralized allocation puts too much of the fixed R&D resource into entry-driven product-line creation. Planner stationary entry is 53.5% below decentralized entry, and the entry R&D share falls from 26.2% to 5.7%. Product-line-weighted average incumbent innovation rises from 0.0666 to 0.0812, while stationary consumption rises by 6.30%. Within incumbent R&D, the planner shifts effort toward older firms whose accumulated buyer-supplier relationships raise the social value of new product lines. Starting from the decentralized steady state, the planner transition raises consumption-equivalent welfare by 2.78%. Figure A4 reports its entry path.

Table 5: Innovation-Return Gaps and Stationary R&D Allocation

	Baseline DE	Appropriation correction	Congestion correction	Planner
Entry R&D share (%)	26.23	21.83	12.64	5.67
Entry change (%)	0.00	-8.79	-30.59	-53.51
Mean incumbent innovation rate	0.0666	0.0689	0.0747	0.0812
Mean incumbent age (R&D-labor weighted, years)	39.7	47.6	73.7	134.9
Stationary consumption change (%)	0.00	1.49	4.28	6.30

Notes: The two middle economies apply the appropriation and congestion corrections separately. Applying both corrections reproduces the planner's R&D allocation. Entry and consumption changes are relative to the baseline decentralized steady state. Mean incumbent innovation is weighted by product-line density. Mean incumbent age is weighted by R&D labor. Consumption changes compare stationary levels, not transition CE gains; the correction comparisons do not provide additive welfare contributions.

The allocation changes reflect the two return gaps described in Section 4.3. Because new product lines created by incumbents inherit existing trading relationships while entrants begin with initial relationships, the two activities differ in both their private returns and their social benefits. In the estimated economy, the planner devotes less R&D labor to entry, leaving more available for incumbent expansion. It also reallocates incumbent R&D toward older firms with accumulated trading relationships. Entry determines the R&D resources left for incumbents, while the planner's value profile guides their allocation across ages.

The product-line-weighted average innovation rate reflects both age-specific choices and the resulting stationary distribution. Transition welfare additionally accounts for the adjustment of the network inherited from the decentralized steady state, whereas the stationary consumption comparison measures the difference between the two long-run allocations.

5.2.2 Innovation-Return Gaps and R&D Allocation

We examine how the two innovation-return gaps in Table 2 shape R&D allocation by correcting them separately and jointly. The appropriation correction modifies fixed transfers so that suppliers receive all bilateral operating surplus. The congestion correction introduces a common levy on active product lines that incorporates the matching-stock term into private innovation values, with the proceeds rebated lump-sum to households. Applied together, these corrections reproduce the planner's R&D allocation. Each economy is solved for its own stationary allocation and production network, holding the estimated technology and preference parameters fixed. Appendix F.11 provides the construction.

Table 5 shows that the appropriation correction shifts R&D from entry toward incumbent expansion. The entry R&D share falls from 26.23% to 21.83%. Within incumbents, effort also shifts toward older firms with accumulated trading relationships, raising the R&D-labor-weighted mean age from 39.7 to 47.6 years. The age-dependent payoff ratio in (25) explains the force behind this reallocation. Full surplus appropriation removes the relative underweighting of returns from established buyer relationships; the equilibrium value profile then directs more R&D toward older incumbents.

Correcting matching-stock congestion alone also shifts R&D toward incumbent expansion and older firms. The entry R&D share falls to 12.64%, and the R&D-labor-weighted mean incumbent age rises to 73.7 years. The common current congestion cost makes the benefits of established trading relationships more important in the valuation of product-line creation. In this economy, its capitalization into innovation values favors expansion through established relationships.

The two corrections are complementary in reallocating R&D away from entry. Applied separately from the decentralized baseline, the appropriation and congestion corrections reduce the entry R&D share by 4.41 and 13.60 percentage points, respectively. Their combined effect is a 20.56-percentage-point reduction, exceeding the sum of the separate effects by 2.56 percentage points. Equivalently, correcting appropriation reduces the entry share by 6.97 percentage points once congestion is priced, compared with 4.41 percentage points when it remains unpriced. Correcting one return gap therefore strengthens the allocation response to correcting the other.

This complementarity also appears in the allocation of incumbent R&D across firm ages. Correcting appropriation raises the R&D-labor-weighted mean age by 7.9 years when congestion is unpriced, but by 61.2 years when it is priced.

Figure 4 compares age profiles across the four stationary economies. In Panel A, innovation rates per product line are lower at the youngest ages and higher at mature ages under the planner than in the decentralized equilibrium. In Panel B, a larger share of total incumbent R&D labor is allocated to older firms under the planner. This age distribution reflects both R&D labor per product line and the number of product lines owned by firms at each age.

5.2.3 Welfare Gains by R&D Allocation Margin

We next quantify how much of the planner's welfare gain comes from reallocating R&D between entry and incumbent expansion, and how much comes from reallocating incumbent R&D across firm ages. These are the two allocation margins characterized in Section 4.1. Each margin reflects the combined effects of the private–social return wedges.

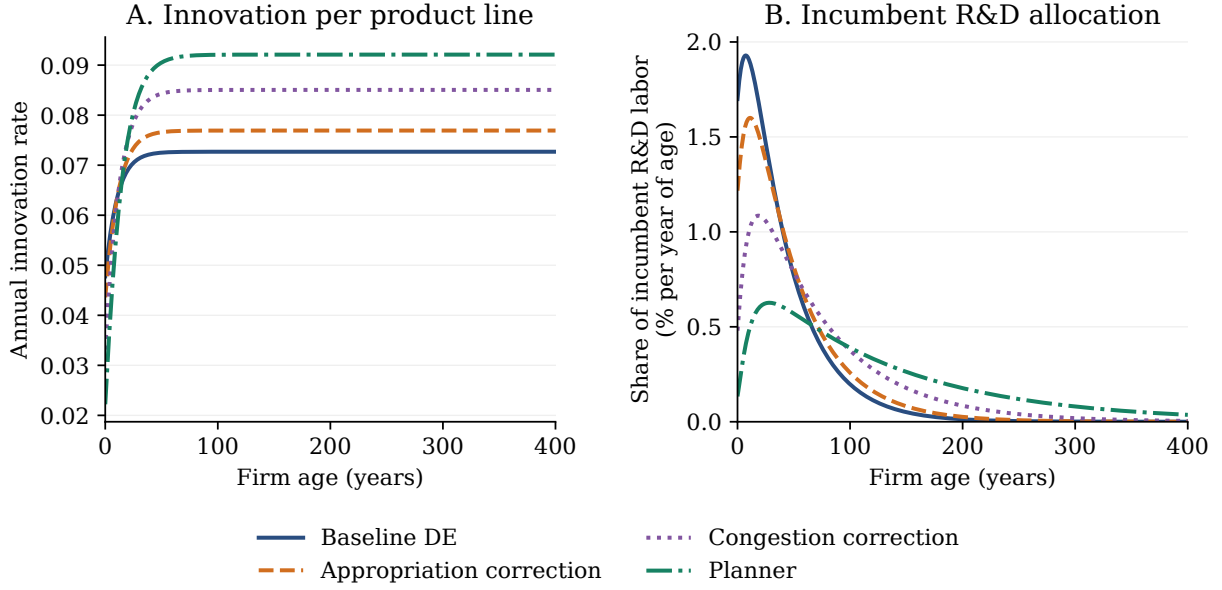


Figure 4: Innovation and R&D Allocation by Firm Age

Notes: The curves correspond to the four economies in Table 5. Panel A shows the annual innovation rate per product line. Panel B shows the distribution of incumbent R&D labor across firm ages, expressed as a percentage of total incumbent R&D labor per year of firm age. The distribution is normalized over all ages, including ages beyond the displayed range.

Table 6: Welfare Gains from R&D Reallocation

Allocation margin	Contribution ($100 \times \log CE$)	Allocation share (%)
Entry	1.801	66.4
Incumbent age	0.912	33.6
Reference path	0.027	—
Total	2.740	—

Notes: Contributions are additive in log CE units and multiplied by 100 for display. Shares divide each Shapley contribution by the sum of the entry and incumbent-age contributions. The reference-path term records the small difference due to numerical approximation of the decentralized reference path, described in Appendix H.3. The total log CE gain corresponds to a 2.78% consumption-equivalent gain.

Starting from the same decentralized steady state, we adopt the planner's allocation along these two dimensions, separately and together. For each dimension, we calculate the welfare gain from changing that allocation when the other follows the decentralized allocation and when it follows the planner's path, and average the two gains. Appendix H.3 provides the construction of this Shapley decomposition.

Table 6 shows that the entry margin contributes 66.4% of the combined allocation gain and the incumbent-age margin contributes 33.6%. Reducing entry releases R&D labor

for incumbent expansion, while reallocating that labor across ages directs innovation toward older firms with accumulated buyer-supplier relationships. Each contribution incorporates the changes in innovation and production-network outcomes induced by the corresponding reallocation.

To illustrate, consider the transition that adopts the planner's entry path while retaining the decentralized allocation of R&D across incumbent ages. At each date, labor used for entry determines how much R&D labor remains for incumbents. This remaining labor is distributed across firm ages using the decentralized conditional shares. Innovation rates at each age are then determined by the labor allocated to that age group and the product lines it owns. As the product-line distribution evolves, innovation rates adjust to preserve the prescribed labor shares, and the network, production costs, and consumption respond accordingly. The two allocation changes therefore interact: the welfare gain from reallocating incumbent R&D across ages depends on how much labor entry leaves available, while the gain from reducing entry depends on how that released labor is allocated.

5.3 Uniform Subsidy to Incumbent R&D

We next ask how much of the planner's welfare gain can be achieved through a uniform subsidy to incumbent R&D. The government reimburses a constant fraction of incumbent R&D expenditure, leaves entry R&D unsubsidized, and finances the subsidy with a lump-sum tax on the household. All firms become eligible for support of product-line expansion after entry. We select the subsidy rate that maximizes welfare along the decentralized transition from the initial steady state. The welfare-maximizing subsidy is 53.6% of incumbent R&D costs. It raises consumption-equivalent welfare by 2.29%, capturing 82.5% of the planner gain in log CE units. Appendix Figure A5 compares consumption along the planner and subsidy transitions.

The subsidy changes both margins of R&D allocation. At a given wage, it raises incumbents' demand for research labor. Competition for the fixed R&D resource raises the research wage and makes unsubsidized entry more costly, shifting labor toward incumbent product-line expansion, which uses established buyer-supplier relationships. Greater incumbent expansion allows firms to accumulate more product lines and employ more R&D workers as they age. Although the subsidy rate is the same for all incumbents, it also shifts incumbent R&D toward older firms with accumulated trading relationships. This shift is in the direction of the welfare-improving reallocation identified in Proposition 2.

These equilibrium responses allow a uniform subsidy to affect both R&D allocation margins without conditioning support on firm age. Within the incumbent sector, the allo-

cation of research labor across ages reflects firms' innovation choices and the accumulation of product lines. The welfare decomposition highlights the quantitative importance of this margin, attributing 33.6% of the combined allocation gain to reallocating incumbent R&D across ages.

With fixed R&D labor supply and lump-sum financing, a constant tax on entry R&D can implement the same transition allocation and welfare gain. The optimal subsidy is equivalent to a 115.7% entry tax. Appendix H.4 gives the policy conditions and the equivalence.

6 Conclusion

This paper studies how production networks shape incentives for innovation by incumbent firms and for firm entry. The empirical evidence shows that firms accumulate buyer-supplier relationships over the life cycle, that older firms tend to trade with older partners, and that firms whose trading partners file more patent applications also file more themselves. Guided by these facts, we develop a dynamic model in which incumbent R&D creates new product lines that inherit existing trading relationships for production and sales. Innovating firms capture only part of the surplus their new products generate for buyers and do not internalize how product-line creation reduces other firms' matching opportunities. Our theoretical analysis establishes that a stationary decentralized equilibrium with accumulating trading relationships admits small, welfare-improving reallocations of R&D from entry to incumbent expansion and, within incumbents, from younger to older firms.

In the estimated economy, R&D is tilted too far toward entry and away from incumbent expansion using established trading relationships. The planner reduces stationary entry by 53.5%, and the transition to the planner allocation raises consumption-equivalent welfare by 2.78%. The entry margin accounts for 66.4% of the combined allocation gain, with the remaining 33.6% coming from reallocating incumbent R&D toward older firms with more accumulated trading relationships. Corrections to surplus appropriation and matching-stock congestion are complementary in reallocating R&D from entry toward incumbent expansion.

A welfare-maximizing uniform subsidy to incumbent R&D captures most of the planner's transition welfare gain by reallocating R&D from entry toward incumbent expansion. These findings highlight the importance for innovation policy of the trading relationships that new products inherit. Production-network accumulation shapes the efficient allocation of R&D by changing the gap between private and social innovation returns across firms.

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Appendix for “Production Networks and R&D Allocation”

This appendix is organized as follows. Appendix A reports additional empirical evidence, and Appendix B describes the estimation targets. Appendix C reports model-implied life-cycle network exposure, while Appendix D derives the closed-form stationary densities. Appendices E–F provide the decentralized and planner derivations, Appendix G describes the numerical implementation and accuracy checks, and Appendix H reports the allocation-margin counterfactuals and policy exercise underlying the welfare results.

A Additional Empirical Evidence

This section supplements the empirical evidence in Section 2 with relationship-survival profiles, cumulative network growth following initial observation, and additional specifications of the partner-patenting regressions.

A.1 Empirical Link Survival

Figure A1 shows the survival profiles of completed buyer-supplier relationship spells. We split a pair’s history at gaps in consecutive annual observations and retain spells that begin more than one year after both firms first appear and end more than one year before both firms last appear in the 1998–2019 sample. The left panel uses spells with manufacturing suppliers and groups them by the supplier’s establishment age at the start; the right uses spells with manufacturing buyers and groups them by the buyer’s establishment age at the start. The other endpoint is unrestricted by industry. The plotted shares are conditional on observing both ends of a spell. The endpoint-survival moments used to calibrate δ_M are described separately in Appendix B.4.

A.2 Life-Cycle Network Profiles

For Figure 1, let K_{it}^s be firm i ’s recorded number of distinct partners of type $s \in \{S, B\}$ in year t . We estimate separately for each partner type

$$\log(1 + K_{it}^s) = \alpha_i^s + \mu_{j(i,t),t}^s + \sum_{a=0}^{49} \beta_a^s \mathbf{1}\{A_{it} = a\} + X'_{it} \gamma^s + \varepsilon_{it}^s,$$

where A_{it} is establishment age, $j(i, t)$ is industry, and X_{it} contains log employment, log capital, and log sales per worker. The regressions jointly estimate age coefficients and

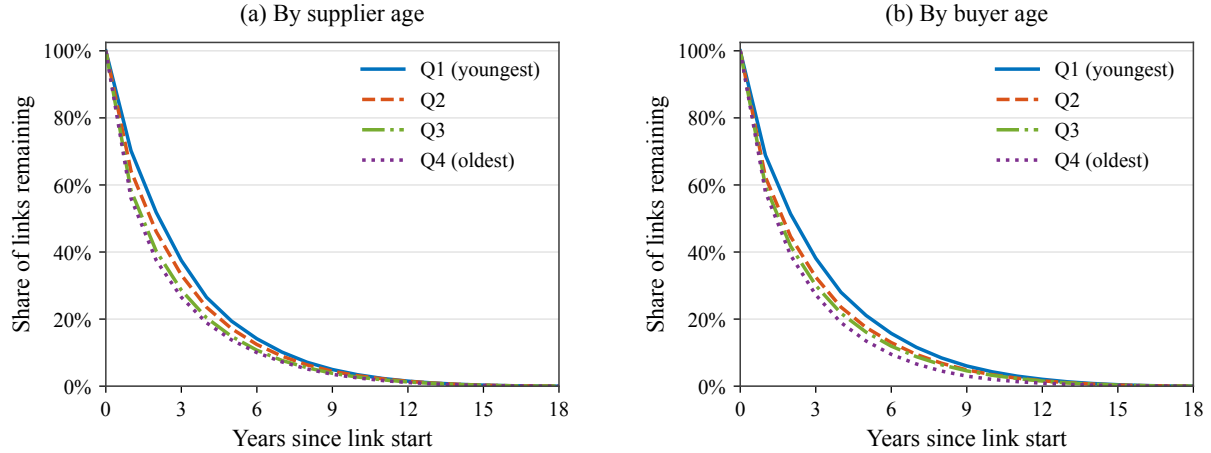


Figure A1: Survival of Completed Trading Relationships

Notes: The figure shows the share of relationships remaining at each elapsed year after their first observation, among spells whose starts and ends are observed within the sample. The left panel groups relationships by the supplier's age at the start of the spell; the right groups them by the buyer's age. Q1 is the youngest quartile and Q4 the oldest. The sample covers 1998–2019.

controls on manufacturing firm-years with nonmissing regression variables; zero recorded partner counts are included. Standard errors are clustered by firm.

Ages 50 and above share a common age effect, normalized to zero. This mature-age restriction anchors the age trend in the presence of firm and time effects, since firm age equals calendar year minus establishment year. The coefficients describe conditional differences in recorded relationships and can also reflect age-varying reporting coverage.

We follow manufacturing firms established in 1998–2004 and observed at age zero or one.¹⁶ For firm i and partner type $s \in \{S, B\}$, let $K_i^s(a)$ denote the number of distinct partners observed at age a and $K_{i,\text{init}}^s$ the count at the first observation in this initial window. We calculate

$$D_i^s(a) = \log(1 + K_i^s(a)) - \log(1 + K_{i,\text{init}}^s).$$

Figure A2 plots the mean of $D_i^s(a)$ among firms observed at each age, pooling the seven establishment cohorts. Firm-years without a matching transaction record have zero recorded partners and are included. There are 14,410 firms at age 1 and 8,070 at age 15. Mean cumulative changes rise with age, reaching 0.93 for suppliers and 0.88 for buyers at age 15.

The profiles before age 15 are nearly unchanged when the sample is restricted to firms also observed at age 15. These profiles are unadjusted for calendar-year and other fixed

¹⁶Age is the TDB record year minus the establishment year, with raw age -1 assigned to zero. If both record years assigned to age zero are observed, the earlier record supplies the initial value.

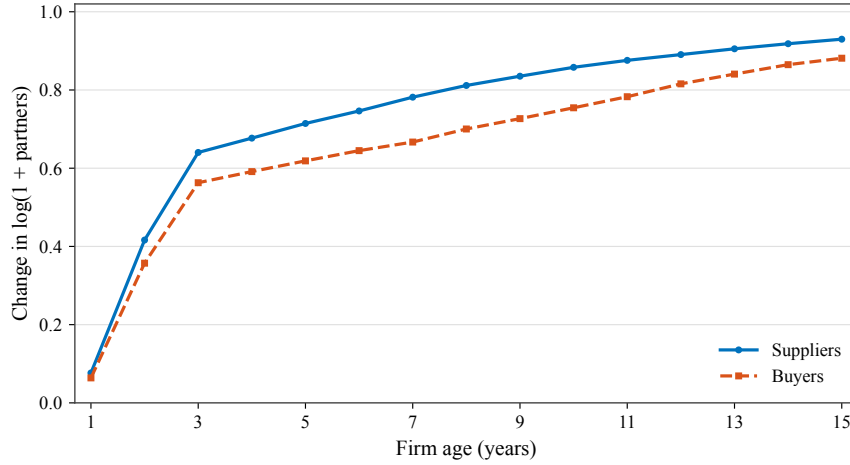


Figure A2: Network Accumulation from Initial Observation

Notes: Mean within-firm changes in $\log(1 + \text{distinct partners})$ from the first observation at age zero or one, pooling manufacturing firms established in 1998–2004. Zero recorded counts are included. Points weight firms observed at each age equally and are unadjusted for fixed effects. Age is years since establishment, measured using the TDB record year; Appendix A.2 describes the cohort construction.

effects, so their age pattern can also reflect calendar-time changes and the composition of observed firms.

A.3 Alternative Specifications for Trading-Partner Patenting

Table A1 compares the joint IV estimates under the baseline and four alternative specifications. Column (1) reproduces the baseline in column (6) of Table 1.

Column (2) examines whether the relationship extends beyond patents that directly cite trading partners. We exclude patents citing a patent held by a trading partner in the citing year, considering both directions of trade, and reconstruct patent counts and patent-class shares. Both partner-patenting coefficients remain positive and statistically significant.

Foreign technological developments may affect both a firm and its trading partners directly. Column (3) therefore controls for the firm’s own foreign-patent exposure, defined as current-year foreign triadic patent families by class weighted by the firm’s previous-year patent-class shares. Column (4) combines this control with the citation-related patent exclusions. The buyer coefficient remains positive and significant across these adjustments. The supplier coefficient remains positive, but is not statistically significant when both adjustments are applied.

Column (5) accounts for common industry shocks by replacing year fixed effects in the baseline specification with industry-by-year fixed effects while retaining firm fixed effects. The supplier and buyer coefficients are smaller than in the baseline but remain

Table A1: Trading-Partner Patenting and Own Patenting: Alternative IV Specifications

	Joint IV				
	(1)	(2)	(3)	(4)	(5)
Supplier patenting per supplier	0.074*** (0.026)	0.057** (0.027)	0.057** (0.023)	0.037 (0.024)	0.059** (0.026)
Buyer patenting per buyer	0.173*** (0.041)	0.182*** (0.041)	0.137*** (0.035)	0.133*** (0.036)	0.119*** (0.038)
Exclude citing patents	No	Yes	No	Yes	No
Own foreign-patent exposure	No	No	Yes	Yes	No
Firm FE	Yes	Yes	Yes	Yes	Yes
Time FE	Year	Year	Year	Year	Industry × year
First-stage F	55.3	56.9	65.9	67.1	55.1
Observations	50,918	48,884	46,720	44,550	49,357

Notes: The dependent variable is log patent applications. All columns jointly include log mean supplier and buyer patenting, using partners identified in the previous year's network. Instruments are log foreign-patent exposures based on lagged partner patent-class shares. Column (1) reproduces column (6) of Table 1. Columns (2) and (4) exclude patents citing a patent held by a trading partner in the citing year, considering both trade directions. Columns (3) and (4) control for log own foreign-patent exposure in both IV stages. Samples require positive values for all variables entering in logs, including own foreign-patent exposure in columns (3) and (4). The source panel covers 1996–2019. Standard errors clustered by industry × year are in parentheses. First-stage F reports the Kleibergen–Paap rk Wald F-statistic. Significance levels are * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

positive and statistically significant. Because the estimation samples differ across columns, coefficient changes may also reflect changes in sample composition.

B Estimation Targets and Calibration

This section describes the empirical targets, model counterparts, and auxiliary calibrations used in Section 5.

Under quadratic entry costs and constant returns in matching, multiplying entry by c and rescaling the R&D parameters as $\varphi \mapsto c\varphi$ and $\varphi_E \mapsto c^2\varphi_E$ preserves innovation rates and R&D labor allocations, with ζ and ζ_0 unchanged. We fix φ_E during estimation. For reporting, we choose $c = 1/E^{DE}$, where E^{DE} is entry in the estimated decentralized steady state before rescaling. The reported efficiencies are therefore φ/E^{DE} and $\varphi_E/(E^{DE})^2$, denoted simply φ and φ_E in the parameter table. Decentralized steady-state entry equals one in these units, which we also use for all policy comparisons.

B.1 Buyer-Supplier Links per Firm

The target for average buyer-supplier links per firm is denoted L^{data} . It is based on the manufacturing JP MOPS sample linked to TDB's COSMOS2 business-to-business transaction data in the February 2022 working-paper version of Imani and Ohyama (2022). Table 1 of that version reports mean supplier and customer (buyer) counts of 28.0 and 26.6 firms, respectively, for Japanese manufacturing firms with establishments in Japan and at least 30 employees. Averaging the reported supplier and buyer counts gives $L^{\text{data}} = (28.0 + 26.6)/2 = 27.3$.

The model counterpart is mean link exposure per firm, obtained by integrating the matched-product density over supplier and buyer ages and dividing by the stationary firm mass $N_f = E/\delta_F$,

$$L^{\text{model}} = \frac{\int \int m(a_s, a_b) f(a_b) da_s da_b}{N_f}.$$

The factor $f(a_b)$ converts the per-buyer-product-line object $m(a_s, a_b)$ into aggregate buyer-supplier links attached to buyer product lines. The model counts relationships at the product-line level rather than at the firm-pair level. It therefore treats two product lines linked to the same trading partner as two units of network exposure, whereas the data would count them as one distinct firm partner. For this reason, the links-per-firm target disciplines the overall scale of network exposure rather than the full distribution of unique firm-level trading partners, primarily through ζ and ζ_0 .

B.2 Life-Cycle Network Coefficients

We construct the life-cycle targets from TDB C2TRD transaction records linked to firm characteristics. For manufacturing firms over 1998–2019, we count distinct suppliers and buyers and measure firm age from establishment. Firms aged 50 and above form a single reference group. We average link counts within age-group, year, prefecture, and industry cells. Separately for suppliers and buyers, we regress $\log(1 + \text{cell mean degree})$ on firm-age indicators and year, prefecture, and industry fixed effects, weighting cells by their number of firm-year observations. At each age $a = 1, \dots, 10$, we take the arithmetic mean of the estimated coefficients, $\bar{\beta}_a^{\text{data}} = (\beta_{S,a}^{\text{data}} + \beta_{B,a}^{\text{data}})/2$, to form ten targets.

The stationary firm-age density is $f_F(a) = E \exp(-\delta_F a)$, while $f(a)$ counts product lines. Hence mean product lines per firm are $\bar{n}(a) = f(a)/f_F(a)$. Supplier and buyer exposure

per product line are

$$L^S(a) = \int m(a_s, a) da_s, \quad L^B(a) = \frac{\int m(a, a_b) f(a_b) da_b}{f(a)}.$$

To construct the model age coefficients, we convert exposure per product line to exposure per firm, $\tilde{L}^s(a) = \bar{n}(a)L^s(a)$, for $s \in \{S, B\}$. The model coefficients for ages $a = 1, \dots, 10$ are

$$\beta_{s,a}^{\text{model}} = \log(1 + \tilde{L}^s(a)) - \log \left(1 + \frac{\int_{50}^{\infty} \tilde{L}^s(\tilde{a}) f_F(\tilde{a}) d\tilde{a}}{\int_{50}^{\infty} f_F(\tilde{a}) d\tilde{a}} \right).$$

The numerical implementation interpolates exposure to integer ages and pools mature exposure using firm-mass weights before taking logs. The model generates the same coefficients for suppliers and buyers, denoted β_a^{model} . The criterion assigns weight 1/10 to each squared difference from the average data coefficient.

B.3 Partner-Patenting Regression

The regression target is the arithmetic mean of the supplier and buyer coefficients in the joint IV specification of Table 1 and is approximately 0.124. The model counterpart is the arithmetic mean of the supplier and buyer coefficients from an auxiliary OLS regression with firm and year effects. The regression relates log own innovation intensity per product line, λ , to the logs of mean supplier and buyer innovation intensities.

We generate the auxiliary panel by simulating firm and product-line exit, innovation, matching, relationship destruction, and inheritance as specified in the stationary model. Supplier and buyer groups are defined by distinct firm partners in the previous year's network. We deduplicate product-line links before averaging partners' current-year innovation intensities. Realized innovation counts determine inclusion in the positive-patent sample, which requires positive own counts and positive total counts in each partner group.

For each candidate value of incumbent R&D efficiency, we reestimate the matching parameters to fit the network targets. We use the same five random seeds across candidates to reduce simulation noise in their comparison, and select the efficiency value at which the mean of the simulated supplier and buyer regression coefficients matches the empirical target. Validation uses ten additional independent seeds, unused in estimation. Each validation run simulates 30,000 firms for 24 observed years after a ten-year burn-in, with within-year time steps of 0.05 years.

B.4 Calibration of Link Destruction

Let S_h denote the share of directed relationships observed at t that are also observed at $t + h$, conditional on both endpoint firms being observed at $t + h$. This measure compares the two observation dates without requiring uninterrupted observations between them. We use links with at least one manufacturing endpoint. The two- and three-year survival rates are $S_2 = 0.8590$ and $S_3 = 0.8122$. Their implied annual hazards are $-\log(S_2)/2 \simeq 0.07598$ and $-\log(S_3)/3 \simeq 0.06933$. We calibrate link destruction to their arithmetic mean,

$$\delta_M = \frac{1}{2} \left[-\frac{\log S_2}{2} - \frac{\log S_3}{3} \right] \simeq 0.07266.$$

The one-year observation horizon gives a higher implied hazard of 0.09234. We use the two longer available horizons to limit sensitivity to short-horizon changes in recorded relationships. These firm-relationship persistence measures serve as a proxy for product-line link destruction. The completed-spell survival profiles in Figure A1 condition on observing both spell endpoints and do not identify the same hazard.

C Model-Implied Life-Cycle Network Exposure

Figure A3 shows a steep early increase in buyer-supplier link exposure per product line and a nearly flat exposure profile among mature firms in the baseline decentralized economy.

D Closed-Form Stationary Densities

Fix an entry flow E and an innovation profile $\lambda(a)$. The stationary product-line density $f(a)$ determines $N = \int f(a)da$, and the matched-product density $m(a_s, a_b)$ then admits a closed-form solution. Here a_s and a_b denote the ages of the supplier and buyer firms associated with a matched product-line pair. Throughout this section, $\delta_M > 0$.

D.1 Product-Line Density

For $a > 0$, the product-line density in (3) satisfies

$$\frac{\partial}{\partial a} f(a) = (\lambda(a) - d) f(a)$$

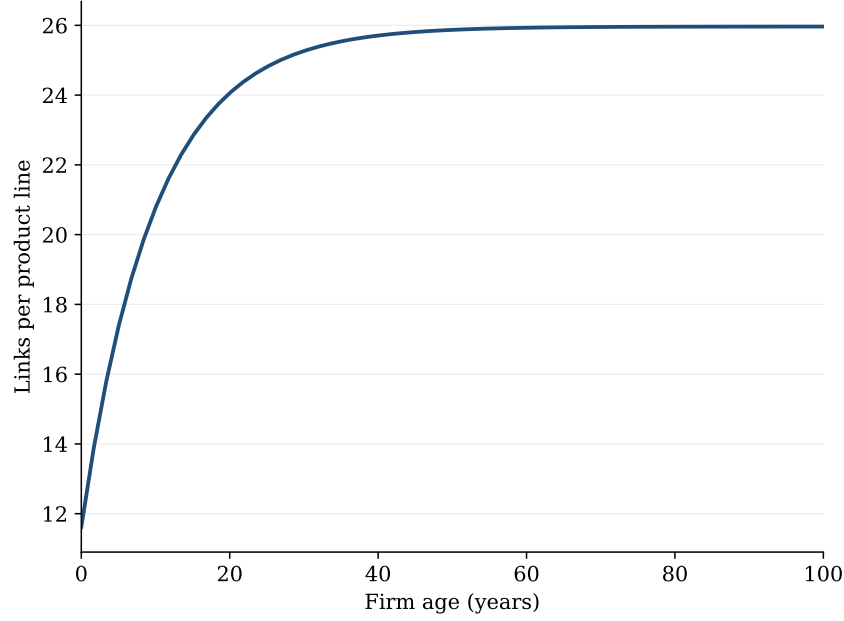


Figure A3: Model-Implied Life-Cycle Network Exposure

Notes: Buyer-supplier link exposure per product line by firm age in the baseline decentralized economy.

with boundary condition $f(0) = E$. Integrating gives

$$f(a) = E \exp \left(\int_0^a \{ \lambda(\tilde{a}) - d \} d\tilde{a} \right). \quad (28)$$

A stationary allocation additionally requires the implied product-line mass to be integrable.

D.2 Matched-Product Density

This subsection derives the closed-form matched-product kernel used in the main text. The key step is to divide out supplier-side product-line mass, which leaves a conditional matching object that evolves only with link destruction and the entrant/interior matching rates.

For the matched-product density, on the interior $a_s > 0$ and $a_b > 0$, (5) is

$$\frac{\partial}{\partial a_s} m(a_s, a_b) + \frac{\partial}{\partial a_b} m(a_s, a_b) = (\lambda(a_s) - d - \delta_M) m(a_s, a_b) + \zeta \frac{f(a_s)}{N},$$

with boundary conditions

$$m(a_s, 0) = \zeta_0 \frac{f(a_s)}{N}, \quad m(0, a_b) = \zeta_0 \frac{f(0)}{N} = \zeta_0 \frac{E}{N}.$$

Entry appears as boundary inflow at age zero; the interior transport equation tracks the evolution of matched product lines after entry. Define the normalized matching kernel by

$$k(a_s, a_b) \equiv \frac{m(a_s, a_b)}{f(a_s)}.$$

Substituting $m(a_s, a_b) = f(a_s)k(a_s, a_b)$ and the product-line density equation into the matching law yields

$$\frac{\partial}{\partial a_s} k(a_s, a_b) + \frac{\partial}{\partial a_b} k(a_s, a_b) = -\delta_M k(a_s, a_b) + \frac{\zeta}{N}. \quad (29)$$

Equation (29) characterizes the normalized conditional matching kernel $k(a_s, a_b)$. The matched-product density is $m(a_s, a_b) = k(a_s, a_b)f(a_s)$, whose tail behavior reflects the supplier-side product-line density $f(a_s)$. The boundary conditions imply

$$k(a_s, 0) = k(0, a_b) = \frac{\zeta_0}{N}.$$

Integrating (29) along the characteristic from the age-zero boundary to (a_s, a_b) , over an age interval of length $\min\{a_s, a_b\}$, gives

$$k(a_s, a_b) = \frac{1}{N} \left[\frac{\zeta}{\delta_M} + \left(\zeta_0 - \frac{\zeta}{\delta_M} \right) \exp(-\delta_M \min\{a_s, a_b\}) \right].$$

Therefore,

$$m(a_s, a_b) = \left[\frac{\zeta}{\delta_M} - \left(\frac{\zeta}{\delta_M} - \zeta_0 \right) \exp(-\delta_M \min\{a_s, a_b\}) \right] \frac{f(a_s)}{N}. \quad (30)$$

The closed form shows that, conditional on f , the innovation profile, supplier-side firm exit, and supplier product-line exit affect $m(a_s, a_b)$ only through the supplier-side mass $f(a_s)$ and its aggregate N . After normalizing by $f(a_s)$, the innovation term and the δ_F and δ_P exit terms in the matched-product law cancel with the corresponding terms in the KFE for $f(a_s)$. The normalized matching kernel depends on ages only through $\min\{a_s, a_b\}$. As the younger firm's age increases, the kernel approaches its long-run level $\zeta/(\delta_M N)$ from its entry-boundary level ζ_0/N at rate δ_M . This closed-form expression also implies the reciprocity relation

$$\frac{m(a_s, a_b)f(a_b)}{f(a_s)} = m(a_b, a_s),$$

which ensures consistency between supplier- and buyer-side normalizations of the same bilateral match mass.

E Decentralized Equilibrium Derivations

This section derives the static allocation, private payoffs, and innovation conditions that close the stationary decentralized equilibrium. We take the age-indexed network (f, m) from Appendix D, with a_s denoting supplier age and a_b buyer age. The density $m(a_s, a_b)$ measures supplier varieties per buyer product line. We normalize final expenditure to $P^F Y = 1$ and suppress the superscript DE . Intermediate inputs are priced at unit cost, final sales carry the markup $\mu^F = \sigma/(\sigma - 1)$, and the maintained transfer rule assigns suppliers a share θ of bilateral operating surplus.

E.1 Unit Costs and Input Demand

For a buyer of age a_b , variable-cost minimization chooses production labor and quantities of linked inputs to minimize

$$wl(a_b) + \int c(a_s)x(a_s, a_b)m(a_s, a_b) da_s$$

subject to producing $x(a_b)$ with the technology in (1). Constant returns imply that minimized variable cost is $c(a_b)x(a_b)$. The CES input aggregator and the normalization of the Cobb–Douglas technology give

$$c(a_b) = w^\beta \left[\int c(a_s)^{1-\sigma} m(a_s, a_b) da_s \right]^{\frac{1-\beta}{1-\sigma}},$$

which is (7). Define the input-expenditure weight per supplier variety by

$$s^M(a_s, a_b) = \frac{c(a_s)^{1-\sigma}}{\int c(\tilde{a}_s)^{1-\sigma} m(\tilde{a}_s, a_b) d\tilde{a}_s}.$$

The conditional factor demands satisfy

$$\begin{aligned} wl(a_b) &= \beta c(a_b)x(a_b), \\ c(a_s)x(a_s, a_b) &= (1 - \beta)c(a_b)x(a_b)s^M(a_s, a_b). \end{aligned} \tag{31}$$

Thus $s^M(a_s, a_b)m(a_s, a_b) da_s$ is the share of intermediate-input expenditure allocated to a supplier-age interval, and $\int s^M(a_s, a_b)m(a_s, a_b) da_s = 1$.

Final demand is $y(a) = Y[p^F(a)/P^F]^{-\sigma}$. Maximizing $[p^F(a) - c(a)]y(a)$ gives $p^F(a) =$

$\mu^F c(a)$, so

$$P^F = \left[\int [\mu^F c(a)]^{1-\sigma} f(a) da \right]^{\frac{1}{1-\sigma}}, \quad R^F(a) \equiv \frac{p^F(a)y(a)}{P^F Y} = \left[\frac{\mu^F c(a)}{P^F} \right]^{1-\sigma}.$$

In particular, $\int R^F(a) f(a) da = 1$.

E.2 Market Clearing, Transfers, and Private Payoffs

Let $C(a) = c(a)x(a)/(P^F Y)$ denote variable production cost per product line in final-expenditure units. Goods-market clearing for suppliers of age a_s is

$$x(a_s)f(a_s) = y(a_s)f(a_s) + \int x(a_s, a_b)m(a_s, a_b)f(a_b) da_b.$$

Multiply by $c(a_s)/(P^F Y f(a_s))$ and use the conditional input demands in (31). Reciprocity, $m(a_s, a_b)f(a_b)/f(a_s) = m(a_b, a_s)$, then yields

$$R^M(a_s) = (1 - \beta) \int C(a_b)s^M(a_s, a_b)m(a_b, a_s) da_b, \\ C(a_s) = \frac{R^F(a_s)}{\mu^F} + R^M(a_s).$$

The first equation sums intermediate sales over the buyer varieties attached to one supplier line. The second recovers the cost of final sales by removing the final markup; intermediate sales are already priced at cost. Conditional on c and hence on (s^M, R^F) , these equations form a linear recursion for C .

Integrating over supplier ages, applying reciprocity and the normalization of s^M , gives

$$\int R^M(a)f(a) da = (1 - \beta) \int C(a)f(a) da, \quad \beta \int C(a)f(a) da = \frac{1}{\mu^F}.$$

The production-labor demand in (31) and the unit labor endowment imply

$$w = \beta \int C(a)f(a) da = \frac{1}{\mu^F}.$$

At fixed buyer output and input prices, differentiating the buyer's minimized variable cost with respect to access to an additional supplier variety gives the marginal cost saving,

in final-expenditure units,

$$S(a_s, a_b) = \frac{1 - \beta}{\sigma - 1} C(a_b) s^M(a_s, a_b).$$

The maintained contract sets the transfer from buyer to supplier to $\theta S(a_s, a_b)$. Hence receipts and payments per product line are

$$\begin{aligned} \theta \int S(a, a_b) m(a_b, a) da_b &= \frac{\theta}{\sigma - 1} R^M(a), \\ \theta \int S(a_s, a) m(a_s, a) da_s &= \frac{\theta(1 - \beta)}{\sigma - 1} C(a). \end{aligned}$$

Intermediate sales earn no variable margin. Adding final-sales profit and net transfers therefore gives, for any fixed supplier share θ ,

$$\begin{aligned} \pi_\theta(a) &= \frac{1}{\sigma} R^F(a) + \frac{\theta}{\sigma - 1} [R^M(a) - (1 - \beta)C(a)] \\ &= (1 - \theta) \frac{R^F(a)}{\sigma} + \theta \frac{\beta C(a)}{\sigma - 1}. \end{aligned} \tag{32}$$

The second equality uses $R^M = C - R^F/\mu^F$. At a given network, this expression is a weighted average of final-sales profit, $R^F(a)/\sigma$, and production-labor expenditure scaled by $1/(\sigma - 1)$, $\beta C(a)/(\sigma - 1)$. The weights are $1 - \theta$ and θ . These are the payoffs at zero and full supplier appropriation, respectively, after netting transfers to and from trading partners. We write this payoff as $\pi(a)$ when the contract share is understood. Transfers cancel upon aggregation over product lines, although they affect private innovation incentives.

E.3 Firm Value, Innovation, and Entry

All product lines within an age- a firm have the same per-line payoff and partner-age distributions. Incumbent innovation adds a line with the same inherited relationships. Payoffs, innovation costs, and product-line event rates scale with the number of lines, so firm value takes the additive form $V^F(n, a) = nV(a)$, with $V^F(0, a) = 0$. A product-line creation raises firm value by $V(a)$, while a product-line exit lowers it by the same amount.

Substituting these increments into the firm Bellman equation (13) and dividing by n gives

$$(\rho + \delta_F)V(a) = \pi(a) + V_a(a) - \delta_P V(a) + \max_{\lambda \geq 0} \{\lambda V(a) - w_H h_H(\lambda)\}.$$

With $d = \delta_F + \delta_P$, this is the product-line HJB (14). The innovation cost $h_H(\lambda) = \lambda^\gamma/\varphi$ is

strictly convex. For a positive value, the first-order condition and policy are

$$V(a) = w_H h'_H(\lambda(a)), \quad \lambda(a) = \left[\frac{\varphi V(a)}{\gamma w_H} \right]^{\frac{1}{\gamma-1}}.$$

Each entrant starts with one age-zero product line, including the initial relationships specified by the matching boundaries. Its value is therefore $V(0)$. With $H_E(E) = E^{\gamma_E}/\varphi_E$, free entry satisfies

$$E \geq 0, \quad w_H H'_E(E) \geq V(0), \quad E[w_H H'_E(E) - V(0)] = 0.$$

At an interior solution,

$$V(0) = w_H \frac{\gamma_E}{\varphi_E} E^{\gamma_E-1}, \quad E = \left[\frac{\varphi_E V(0)}{\gamma_E w_H} \right]^{\frac{1}{\gamma_E-1}}.$$

Entry and incumbent innovation compete for the same unit endowment of research labor, so

$$1 = \frac{E^{\gamma_E}}{\varphi_E} + \int \frac{\lambda(a)^\gamma}{\varphi} f(a) da.$$

Together with the stationary densities in Appendix D and the static allocation above, these conditions close the equilibrium defined in Section 3. Appendix G.1 describes the numerical iteration that solves them jointly.

F Social Values and R&D Misallocation

We derive the social values of product lines and trading relationships in Proposition 1 and the resulting optimal R&D and entry conditions. We first solve the static allocation to measure current benefits, then account for how innovation, entry, and the evolution of trading relationships affect future consumption. We then characterize age differences in private and social innovation returns and prove that reallocating R&D labor can improve welfare. Finally, we describe the separate and joint corrections to the two return gaps used in the quantitative analysis. Throughout this appendix, the first argument of $x(\cdot, \cdot)$, $m(\cdot, \cdot)$, and $V^M(\cdot, \cdot)$ is the supplier age and the second argument is the buyer age; when both sides appear, we write them as a_s and a_b .

F.1 Objective and Constraints

The social planner maximizes $\int_0^\infty e^{-\rho t} \log Y(t) dt$ subject to the production technology (20), the goods market clearing condition

$$x(a_s) f(a_s) = y(a_s) f(a_s) + \int x(a_s, a_b) m(a_s, a_b) f(a_b) da_b, \quad (33)$$

the laws of motion (3), (5), the entry boundary condition (4), the entrant boundary conditions for $m(a_s, a_b)$, and the factor constraints (22), (16). In (33), a_s is in the supplier position and the integral runs over all buyer ages a_b .

F.2 Lagrangian and Hamiltonian

The intratemporal Lagrangian characterizes the allocation of goods and production labor at a given network state, while the Hamiltonian incorporates the effects of R&D and entry on future product lines and trading relationships.

With states (f, m) fixed, the intratemporal Lagrangian is

$$\begin{aligned} \mathcal{L}_0^{SP} = & \log Y \\ & + \int \chi(a_b) \left[\frac{l(a_b)^\beta J(a_b)^{\frac{\sigma(1-\beta)}{\sigma-1}}}{\beta^\beta (1-\beta)^{1-\beta}} - x(a_b) \right] f(a_b) da_b \\ & + \int p^{SP}(a_s) \left[x(a_s) f(a_s) - y(a_s) f(a_s) - \int x(a_s, a_b) m(a_s, a_b) f(a_b) da_b \right] da_s \\ & + v_L \left[1 - \int l(a) f(a) da \right], \end{aligned}$$

where $J(a_b) \equiv \int x(a_s, a_b)^{(\sigma-1)/\sigma} m(a_s, a_b) da_s$ is the CES intermediate bundle of buyer a_b , $\chi(a)$ is the shadow value of relaxing buyer a 's production constraint, $p^{SP}(a)$ is the planner's shadow-price multiplier for good a rather than a decentralized transaction price, and v_L is the shadow wage of production labor.

Imposing the entry boundary condition for f and the entrant boundary conditions for $m(a_s, a_b)$ separately on $a_s = 0$ and $a_b = 0$, the Hamiltonian over the interior states is

$$\begin{aligned} \mathcal{H}^{SP} = & \mathcal{L}_0^{SP} + \int \xi_f(a) \left[-\frac{\partial f}{\partial a} + (\lambda(a) - d) f(a) \right] da \\ & + \iint \xi_m(a_s, a_b) \left[-\frac{\partial m}{\partial a_s} - \frac{\partial m}{\partial a_b} \right] da_s da_b \end{aligned}$$

$$+ (\lambda(a_s) - d - \delta_M) m(a_s, a_b) + \zeta \frac{f(a_s)}{N} \Big] da_s da_b \\ + \xi_\lambda \left[1 - H_E(E) - \int h_H(\lambda(a)) f(a) da \right].$$

Here ξ_f and ξ_m are the current-value costates associated with f and m , respectively, and ξ_λ is the multiplier on the R&D resource constraint. The boundary conditions are $f(0) = E$, $m(a_s, 0) = \zeta_0 f(a_s)/N$, and $m(0, a_b) = \zeta_0 E/N$. The coordinate derivatives below first hold the common denominator N fixed; Section F.7 then adds the effect of changing $N = \int f(a) da$.

F.3 Static First-Order Conditions

FOC w.r.t. $y(a)$. Differentiating $\log Y$ gives

$$p^{SP}(a) = \frac{y(a)^{-1/\sigma}}{\int y(\tilde{a})^{(\sigma-1)/\sigma} f(\tilde{a}) d\tilde{a}}.$$

FOC w.r.t. $x(a)$. From the production and resource constraints:

$$\chi(a) = p^{SP}(a). \quad (34)$$

FOC w.r.t. $l(a)$.

$$p^{SP}(a) \beta x(a) = v_L l(a). \quad (35)$$

FOC w.r.t. $x(a_s, a_b)$: CES optimality. Define the input-expenditure weight per supplier variety

$$s(a_s, a_b) \equiv \frac{x(a_s, a_b)^{(\sigma-1)/\sigma}}{J(a_b)}.$$

The normalization is $\int s(a_s, a_b) m(a_s, a_b) da_s = 1$, so $s(a_s, a_b) m(a_s, a_b) da_s$ is the input-expenditure share of suppliers in the age interval $[a_s, a_s + da_s]$. The per-match quantity $x(a_s, a_b)$ enters buyer a_b 's production through $J(a_b)$ and supplier a_s 's resource constraint. The FOC yields

$$p^{SP}(a_b) (1 - \beta) x(a_b) \frac{x(a_s, a_b)^{-1/\sigma}}{J(a_b)} = p^{SP}(a_s). \quad (36)$$

Multiplying both sides by $x(a_s, a_b)$, we obtain the useful static identity:

$$p^{SP}(a_s) x(a_s, a_b) = p^{SP}(a_b) (1 - \beta) x(a_b) s(a_s, a_b). \quad (37)$$

F.4 Cost and Demand Shifters

The static first-order conditions determine the cost and demand shifters used to compute R^{SP} in the planner value equations. From constant-returns cost duality in the static block, the planner's shadow price is proportional to the full marginal cost shifter:

$$p^{SP}(a) \propto c^{SP}(a).$$

To fix units, temporarily write production labor as L . At fixed (f, k) , scaling labor and all goods quantities by the same factor preserves the constant-returns technologies and goods-market clearing. Consequently $Y(L; f, k) = LY(1; f, k)$ and

$$\nu_L = \frac{\partial \log Y(L; f, k)}{\partial L} = \frac{1}{L} = 1 \quad \text{at } L = 1.$$

Unit-cost duality therefore identifies the shadow price $p^{SP}(a)$ with the cost shifter $c^{SP}(a)$ computed at unit production wage.

Substituting this relation into the CES input-demand FOC (36) gives

$$x(a_s, a_b) = (1 - \beta) c^{SP}(a_s)^{-\sigma} c^{SP}(a_b)^{\frac{\sigma - \beta}{1 - \beta}} x(a_b).$$

Inserting this into the production function (20) for buyer a_b and cancelling $x(a_b)$ from both sides yields the cost recursion:

$$c^{SP}(a_b) = \left(\int c^{SP}(a_s)^{1 - \sigma} m(a_s, a_b) da_s \right)^{\frac{1 - \beta}{1 - \sigma}}. \quad (38)$$

Relabeling the left-hand-side age as a , substituting into the planner goods-market clearing condition (33), and defining

$$D^{SP}(a) \equiv x(a) [P^{SP}]^{1 - \sigma} [c^{SP}(a)]^\sigma$$

gives

$$D^{SP}(a) = (1 - \beta) \int [c^{SP}(a_b)]^{\frac{\beta}{1 - \beta}(\sigma - 1)} D^{SP}(a_b) \frac{m(a, a_b) f(a_b)}{f(a)} da_b + 1.$$

From the labor market clearing condition (22):

$$(P^{SP})^{1 - \sigma} = \beta \int c^{SP}(a)^{1 - \sigma} D^{SP}(a) f(a) da,$$

and the CES price index is

$$P^{SP} = \left(\int c^{SP}(a)^{1-\sigma} f(a) da \right)^{\frac{1}{1-\sigma}}. \quad (39)$$

Using the demand-shifter definition above,

$$R^{SP}(a) = c^{SP}(a)x(a) = p^{SP}(a)x(a).$$

These are the social planner analogs of the decentralized static cost and demand objects. Unlike the decentralized contracting block, the planner prices intermediate and final uses at social marginal cost and does not use current fixed transfers to allocate bilateral operating surplus.

F.5 Matched-Product Density Envelope

This subsection derives the net current benefit of an additional trading relationship by subtracting its resource cost from the buyer's gain from input variety. This benefit enters the match-value costate equation and contributes to the social value of innovation through inherited trading relationships.

Define the reduced current return

$$\Psi^{SP}(f, m) \equiv \max_{y, x, l, \{x(a_s, a_b)\}} \mathcal{L}_0^{SP}.$$

By the envelope theorem, holding the optimal static allocations fixed, $m(a_s, a_b)$ enters $\mathcal{L}_0^{SP}$ in exactly two places: (i) buyer a_b 's intermediate bundle $J(a_b)$, and (ii) supplier a_s 's resource constraint.

Gross variety benefit. A marginal increase in the matched-product density $m(a_s, a_b)$ expands buyer a_b 's input variety:

$$\frac{\partial x(a_b)}{\partial m(a_s, a_b)} = \frac{\sigma(1-\beta)}{\sigma-1} x(a_b) s(a_s, a_b).$$

Valued at $p^{SP}(a_b) f(a_b)$ using (34), the gross variety benefit is

$$\text{Variety benefit} = p^{SP}(a_b) f(a_b) \frac{\sigma(1-\beta)}{\sigma-1} x(a_b) s(a_s, a_b).$$

Resource cost. Supplier a_s must deliver $x(a_s, a_b)$ units through this match:

$$\text{Resource cost} = p^{SP}(a_s) x(a_s, a_b) f(a_b).$$

Net flow benefit of a match. Combining the gross variety benefit and the resource cost:

$$\frac{\partial \Psi^{SP}}{\partial m(a_s, a_b)} = f(a_b) \left[p^{SP}(a_b) \frac{\sigma(1-\beta)}{\sigma-1} x(a_b) s(a_s, a_b) - p^{SP}(a_s) x(a_s, a_b) \right].$$

The first term is the gross variety benefit for buyer a_b , and the second is the resource cost for supplier a_s .

Simplification via CES identity. Using the static identity (37) to substitute the resource cost:

$$\frac{\partial \Psi^{SP}}{\partial m(a_s, a_b)} = \frac{1-\beta}{\sigma-1} p^{SP}(a_b) f(a_b) x(a_b) s(a_s, a_b). \quad (40)$$

The factor $\sigma/(\sigma-1) - 1 = 1/(\sigma-1) > 0$ is the net CES variety surplus. This net flow payoff is the primitive source term in the match-value costate equation below. At the planner's shadow prices, the gross variety benefit exceeds the resource cost of supplying the additional input. In the decentralized economy, the supplier receives intermediate-sales revenue and the fixed transfer θS^{DE} , and therefore captures part of the bilateral operating surplus. Private contracts do not, however, price the full planner value of the match portfolio. The social product-line value incorporates this match surplus through the inherited portfolio.

F.6 Product-Line Density Envelope

This subsection derives the direct effect of product-line density on the optimized current payoff; the effect through aggregate matching intensity is derived separately in the next subsection.

$f(a)$ enters $\mathcal{L}_0^{SP}$ through four terms. First, the objective $\log Y$ contributes $\frac{\sigma}{\sigma-1} p^{SP}(a) y(a)$. Second, when age a is the buyer, the resource constraint for intermediate purchases contributes

$$- \int p^{SP}(\tilde{a}) x(\tilde{a}, a) m(\tilde{a}, a) d\tilde{a} = -p^{SP}(a) (1-\beta) x(a),$$

using the aggregate intermediate-use identity implied by (37), the CES share definition, and (33). Third, when age a is the supplier, the resource constraint contributes $p^{SP}(a) [x(a) - y(a)]$.

Fourth, the labor constraint contributes $-v_L l(a) = -p^{SP}(a) \beta x(a)$, using (35). Summing:

$$\begin{aligned} \frac{\partial \Psi^{SP}}{\partial f(a)} &= \frac{\sigma}{\sigma-1} p^{SP}(a) y(a) + p^{SP}(a) [x(a) - y(a)] \\ &\quad - p^{SP}(a) (1 - \beta) x(a) - p^{SP}(a) \beta x(a) \\ &= \frac{p^{SP}(a) y(a)}{\sigma-1}. \end{aligned} \quad (41)$$

F.7 Aggregate Matching-Stock Envelope

The coordinate envelopes above hold the denominator N fixed. Private agents take the aggregate product-line stock as given, while the planner internalizes how an additional product line changes matching intensity with existing counterpart lines through $1/N$. Its log derivative with respect to any age- a product line is

$$\frac{\delta \log(1/N)}{\delta f(a)} = -\frac{1}{N}. \quad (42)$$

Write the normalized match value as $V^M(a_s, a_b) = (\sigma - 1)\xi_m(a_s, a_b)/f(a_b)$. This normalization multiplies the utility-based value per match by $\sigma - 1$. The normalized value exposure to a proportional change in $1/N$ is

$$\begin{aligned} \Omega \equiv \frac{1}{N} &\left[\zeta \iint V^M(a_s, a_b) f(a_s) f(a_b) da_s da_b \right. \\ &\quad \left. + \zeta_0 E \int V^M(a_s, 0) f(a_s) da_s + \zeta_0 E \int V^M(0, a_b) f(a_b) da_b \right]. \end{aligned} \quad (43)$$

The first term is the value exposure of interior match formation; the last two terms are the corresponding entrant-as-buyer and entrant-as-supplier boundary exposures. Combining (42) with (43) gives the product-line matching-stock source

$$\Omega \frac{\delta \log(1/N)}{\delta f(a)} = -\frac{\Omega}{N}. \quad (44)$$

In the product-line-stock specification this source is common across ages, because every product line contributes one unit to N . It is the matching-stock congestion term in (23). Section F.8 derives this term from the Hamiltonian, including the contributions of the matching boundaries.

F.8 Dynamic First-Order Conditions

To value a product line together with its inherited relationships, we separate changes in product-line density from changes in the normalized matching kernel. We then derive the current-value costates and the R&D and entry conditions. Time arguments are suppressed where no ambiguity arises.

F.8.1 Inherited-Portfolio Coordinates

For $f(a_s, t) > 0$, write the existing normalized matching kernel with explicit time dependence and introduce the derivative operators

$$k(a_s, a_b, t) = \frac{m(a_s, a_b, t)}{f(a_s, t)}, \quad \mathcal{D}_a = \partial_t + \partial_{a_s}, \quad \mathcal{D}_{sb} = \partial_t + \partial_{a_s} + \partial_{a_b}.$$

The density f counts product lines by the age of their firm. The normalized kernel k describes the matching state per unit of supplier product-line density. In particular, $k(a_s, a_b)f(a_b)da_b$ is the inherited mass of buyer product lines in age interval da_b attached to one age- a_s supplier product line. The product-line state and the original matched-density law imply

$$\begin{aligned} \mathcal{D}_a f(a) &= (\lambda(a) - d)f(a), \\ \mathcal{D}_{sb} m(a_s, a_b) &= (\lambda(a_s) - d - \delta_M)m(a_s, a_b) + \frac{\zeta f(a_s)}{N}. \end{aligned}$$

As in Section D, dividing by supplier product-line density cancels the common innovation and exit terms, giving the dynamic matching law

$$\mathcal{D}_{sb} k(a_s, a_b) = -\delta_M k(a_s, a_b) + \frac{\zeta}{N}. \quad (45)$$

Thus supplier innovation and exit change the number of inherited portfolios through f . The normalized kernel k evolves through link formation and destruction; the buyer-link mass per supplier product line also depends on the evolving buyer density $f(a_b)$. The boundary conditions transform to

$$f(0) = E, \quad k(a_s, 0) = \frac{\zeta_0 f(a_s)/N}{f(a_s)} = \frac{\zeta_0}{N}, \quad k(0, a_b) = \frac{\zeta_0 E/N}{E} = \frac{\zeta_0}{N}.$$

The expressions at age zero use the positive-entry solution. Entry changes the product-line boundary; the normalized matching boundaries depend on the aggregate stock.

Transforming the costates. Let $q(a)$ and $h(a_s, a_b)$ denote the current-value multipliers of f and k . For arbitrary state variations, $\delta m = f(a_s)\delta k + k\delta f(a_s)$. Substitution into the primitive costate pairing gives

$$\begin{aligned} & \int \xi_f(a)\delta f(a) da + \iint \xi_m(a_s, a_b)\delta m(a_s, a_b) da_s da_b \\ &= \int \left[\xi_f(a) + \int \xi_m(a, a_b)k(a, a_b) da_b \right] \delta f(a) da + \iint \xi_m(a_s, a_b)f(a_s)\delta k(a_s, a_b) da_s da_b. \end{aligned}$$

Equating the coefficients of the two variations yields

$$q(a) = \xi_f(a) + \int \xi_m(a, a_b)k(a, a_b) da_b, \quad h(a_s, a_b) = \xi_m(a_s, a_b)f(a_s).$$

The portfolio contribution in q is the value of replicating the inherited buyer links along with the supplier product line. This transformation preserves the shadow value of every state perturbation and applies along a transition as well as at a stationary allocation.

Static envelopes in portfolio coordinates. Let $u(f, k)$ denote maximized current log consumption in the transformed coordinates, so $u(f, k) = \Psi^{SP}(f, f k)$. The CES factor $1/(\sigma - 1)$ is common across ages and constant over time.

Using the static objects in Section F.4, the chain rule at fixed k combines the product-line envelope in (41) and the matched-density envelope in (40):

$$\begin{aligned} u_f(a) &= \Psi_f^{SP}(a) + \int \Psi_m^{SP}(a, a_b)k(a, a_b) da_b \\ &= \frac{p^{SP}(a)y(a)}{\sigma - 1} + \frac{1}{(\sigma - 1)f(a)} \int (1 - \beta)p^{SP}(a_b)x(a_b)s(a, a_b)m(a, a_b)f(a_b) da_b. \end{aligned} \quad (46)$$

By (37), the integrand's price-quantity term equals $p^{SP}(a)x(a, a_b)$. Reciprocity and goods-market clearing then give

$$\begin{aligned} \frac{1}{f(a)} \int x(a, a_b)m(a, a_b)f(a_b) da_b &= \int x(a, a_b)m(a_b, a) da_b \\ &= x(a) - y(a). \end{aligned}$$

Substituting into (46) makes the aggregation explicit:

$$u_f(a) = \frac{p^{SP}(a)y(a)}{\sigma - 1} + \frac{p^{SP}(a)[x(a) - y(a)]}{\sigma - 1} = \frac{R^{SP}(a)}{\sigma - 1}.$$

At fixed f , $\delta m(a_s, a_b) = f(a_s)\delta k(a_s, a_b)$ instead implies

$$u_k(a_s, a_b) = f(a_s)\Psi_m^{SP}(a_s, a_b) = \frac{1-\beta}{\sigma-1}R^{SP}(a_b)s(a_s, a_b)f(a_s)f(a_b).$$

The current benefit of replicating a product line thus includes its final-consumption contribution and the surplus it supplies through inherited relationships.

F.8.2 Dynamic Value Equations

Hamiltonian variation and age boundaries. The transformed current-value Hamiltonian, including the R&D constraint, is

$$\begin{aligned}\tilde{\mathcal{H}} = & u(f, k) + \xi_\lambda \left[1 - H_E(E) - \int h_H(\lambda(a))f(a) da \right] \\ & + \int q(a)[- \partial_a f(a) + (\lambda(a) - d)f(a)] da \\ & + \iint h(a_s, a_b) \left[-(\partial_{a_s} + \partial_{a_b})k(a_s, a_b) \right. \\ & \quad \left. - \delta_M k(a_s, a_b) + \zeta/N \right] da_s da_b.\end{aligned}$$

Consider variations with fixed initial states and vanishing costate-weighted boundary terms at infinite age and time. Equivalently, the calculation can be made on bounded domains retaining their boundary terms before taking the limit. Integration by parts in time gives $\partial_t q - \rho q$ and $\partial_t h - \rho h$ as the coefficients of the state variations in the discounted integrand. Integration by parts in age similarly yields

$$\begin{aligned}- \int q \partial_a \delta f da &= q(0)\delta E + \int (\partial_a q)\delta f da, \\ - \iint h(\partial_{a_s} + \partial_{a_b})\delta k da_s da_b &= \int h(0, a_b)\delta k(0, a_b) da_b + \int h(a_s, 0)\delta k(a_s, 0) da_s \\ &\quad + \iint (\partial_{a_s} h + \partial_{a_b} h)\delta k da_s da_b.\end{aligned}\tag{47}$$

The two normalized matching boundaries have variations

$$\delta k(0, a_b) = \delta k(a_s, 0) = -\frac{\zeta_0}{N^2}\delta N, \quad \delta N = \int \delta f(a) da.$$

The interior source contributes $-\zeta \iint h(a_s, a_b) da_s da_b \delta N / N^2$. Combining it with the boundary terms in (47) gives

$$-\frac{\Omega}{(\sigma-1)N} = -\frac{\zeta \iint h(a_s, a_b) da_s da_b + \zeta_0 \int [h(a, 0) + h(0, a)] da}{N^2}. \quad (48)$$

Each interior product line changes N by the same amount, so this contribution is common across ages. The two boundary integrals value entrant-as-buyer and entrant-as-supplier links separately; they are not additional direct rewards for choosing E .

Interior costate equations. Collecting the coefficients of δf and δk in the augmented objective gives

$$\begin{aligned} (\rho + d - \lambda(a))q(a) &= \mathcal{D}_a q(a) + u_f(a) - \xi_\lambda h_H(\lambda(a)) - \frac{\Omega}{(\sigma-1)N}, \\ (\rho + \delta_M)h(a_s, a_b) &= \mathcal{D}_{sb} h(a_s, a_b) + u_k(a_s, a_b). \end{aligned} \quad (49)$$

Express the social values and the R&D shadow wage in units that multiply their utility-based counterparts by $\sigma - 1$.

$$\begin{aligned} V^{SP}(a) &= (\sigma-1)q(a), & w_H^{SP} &= (\sigma-1)\xi_\lambda, \\ V^M(a_s, a_b) &= \frac{(\sigma-1)h(a_s, a_b)}{f(a_s)f(a_b)} = \frac{(\sigma-1)\xi_m(a_s, a_b)}{f(a_b)}. \end{aligned}$$

In particular $h(a, 0) = V^M(a, 0)f(a)E/(\sigma-1)$ and $h(0, a) = V^M(0, a)Ef(a)/(\sigma-1)$. Substituting these boundary values into (48) and multiplying by $\sigma - 1$ gives the normalized source $-\Omega/N$ in (43)–(44).

For the match value, the density normalization matters along both age directions:

$$\begin{aligned} \mathcal{D}_{sb}[f(a_s)f(a_b)] &= [\lambda(a_s) + \lambda(a_b) - 2d]f(a_s)f(a_b), \\ \frac{(\sigma-1)\mathcal{D}_{sb}h(a_s, a_b)}{f(a_s)f(a_b)} &= \mathcal{D}_{sb}V^M(a_s, a_b) + [\lambda(a_s) + \lambda(a_b) - 2d]V^M(a_s, a_b). \end{aligned}$$

Dividing the two primitive costate equations by their respective normalizations consequently gives

$$\begin{aligned} (\rho + d - \lambda(a))V^{SP}(a) &= \mathcal{D}_a V^{SP}(a) + R^{SP}(a) \\ &\quad - w_H^{SP} h_H(\lambda(a)) - \frac{\Omega}{N}, \end{aligned} \quad (50)$$

$$\begin{aligned}
& [\rho + 2d + \delta_M - \lambda(a_s) - \lambda(a_b)]V^M(a_s, a_b) \\
& = \mathcal{D}_{sb}V^M(a_s, a_b) + (1 - \beta)R^{SP}(a_b)s(a_s, a_b).
\end{aligned}$$

Here

$$s(a_s, a_b) = \frac{c^{SP}(a_s)^{1-\sigma}}{\int c^{SP}(\tilde{a})^{1-\sigma} m(\tilde{a}, a_b) d\tilde{a}}.$$

The match value $V^M(a_s, a_b)$ is used to evaluate the aggregate matching-stock term $-\Omega/N$ derived in Section F.7. Setting the time derivatives to zero yields Proposition 1.

Evolution of the inherited portfolio. To identify how link formation and destruction enter the product-line value in (50), consider its inherited-relationship component $q(a_s) - \xi_f(a_s) = \int \xi_m(a_s, a_b)k(a_s, a_b) da_b$. The product rule and buyer-age integration by parts give

$$\begin{aligned}
\mathcal{D}_{a_s}[q(a_s) - \xi_f(a_s)] &= \int k(a_s, a_b)\mathcal{D}_{sb}\xi_m(a_s, a_b) da_b \\
&+ \int \xi_m(a_s, a_b) \left[-\delta_M k(a_s, a_b) + \frac{\zeta}{N} \right] da_b + \xi_m(a_s, 0)k(a_s, 0).
\end{aligned}$$

The last two terms capture the portfolio's link evolution and its buyer-age boundary contribution. Through the state law and boundary variations, both enter the derivative of $q - \xi_f$ and hence the full portfolio value in (50), rather than appearing as separate HJB source terms.

F.8.3 Optimal R&D and Entry

Holding the states fixed, the terms multiplying an innovation perturbation are

$$\delta_\lambda \tilde{\mathcal{H}} = \int f(a)[q(a) - \xi_\lambda h'_H(\lambda(a))]\delta\lambda(a) da.$$

For positive innovation the bracket is zero. With $h_H(\lambda) = \lambda^\gamma/\varphi$, multiplication by $\sigma - 1$ gives

$$\lambda^{SP}(a) = \left[\frac{\varphi V^{SP}(a)}{\gamma w_H^{SP}} \right]^{1/(\gamma-1)}.$$

At a zero-innovation corner, the marginal social value is no greater than the marginal resource cost.

For entry, (47) and the entry labor cost give the boundary variation

$$[q(0) - \xi_\lambda H'_E(E)]\delta E.$$

The dependence of the matching boundaries on N has already been included in the matching-stock term in (48). Their normalized values have no remaining direct dependence on E . The interior condition is therefore

$$q(0) = \xi_\lambda H'_E(E).$$

The same condition can be recovered in the original coordinates: since $k(0, a_b) = \zeta_0/N$,

$$q(0) = \xi_f(0) + \frac{\zeta_0}{N} \int \xi_m(0, a_b) da_b.$$

Thus the entrant's initial portfolio of buyer links is included in its product-line value. In normalized units,

$$V^{SP}(0) = w_H^{SP} H'_E(E). \quad (51)$$

The KKT extension is $E \geq 0$, $w_H^{SP} H'_E(E) \geq V^{SP}(0)$, and $E[w_H^{SP} H'_E(E) - V^{SP}(0)] = 0$. For $H_E(E) = E^{\gamma_E}/\varphi_E$ and positive entry,

$$E = \left[\frac{\varphi_E V^{SP}(0)}{\gamma_E w_H^{SP}} \right]^{1/(\gamma_E-1)}. \quad (52)$$

The R&D resource constraint determines the common multiplier w_H^{SP} , linking entry to the resources available for incumbent innovation.

F.9 Surplus Appropriation and Relative Innovation Returns

This subsection derives the age pattern of relative current returns and its connection to dynamic innovation values. It also establishes the full-surplus-appropriation comparison in Section 5.2.2. We retain the baseline marginal prices and first allow the supplier's surplus share θ to vary while holding (f, m) fixed. Transfers affect private profits but leave the static goods allocation unchanged at that state.

Private and social current returns. Appendix E.2 derives the private current return (32) from final-sales profits and net transfers. We restore the superscript DE on its static objects to compare them with the planner's allocation. At the same network state, the planner's current return satisfies $R^{SP}(a) = \mu^F C^{DE}(a) = \sigma C^{DE}(a)/(\sigma - 1)$. This follows by multiplying the decentralized recursion for variable production costs by μ^F . The result is the planner's revenue recursion because relative costs and input shares coincide under the baseline

marginal prices. Substitution into (32) gives

$$\pi_\theta(a) = \frac{1-\theta}{\sigma} R^{F,DE}(a) + \frac{\theta\beta}{\sigma} R^{SP}(a), \quad \pi_1(a) = \frac{\beta}{\sigma} R^{SP}(a). \quad (53)$$

Age profile of relative current returns. Relative final-consumption prices coincide at a common network state, so $R^{F,DE}(a) = R^{SP}(a)/D^{SP}(a)$. Substitution into (53) gives (25). To establish its age profile, consider a stationary network with a positive, finite static solution and $\zeta > \delta_M \zeta_0$, the network-accumulation condition satisfied by the estimate. The closed form in (6) implies, away from the diagonal,

$$\partial_{a_s} k(a_s, a_b) = \frac{\zeta - \delta_M \zeta_0}{N} e^{-\delta_M a_s} \mathbf{1}\{a_b > a_s\}.$$

Using $m(a_s, a_b)f(a_b)/f(a_s) = k(a_s, a_b)f(a_b)$ in the demand recursion gives

$$\begin{aligned} \partial_a D^{SP}(a) &= \frac{(1-\beta)(\zeta - \delta_M \zeta_0)}{N} e^{-\delta_M a} \\ &\times \int_a^\infty [c^{SP}(b)]^{\frac{\beta}{1-\beta}(\sigma-1)} D^{SP}(b) f(b) db \geq 0. \end{aligned}$$

The inequality is strict whenever older counterpart lines have positive mass. Consequently, for $0 \leq \theta < 1$,

$$\partial_a \left(\frac{\pi_\theta(a)}{R^{SP}(a)} \right) = -\frac{1-\theta}{\sigma [D^{SP}(a)]^2} \partial_a D^{SP}(a) \leq 0.$$

Private current returns thus place relatively more weight on entry and young-firm innovation than on innovation using older firms' established buyer relationships.

The same kernel also expands supplier access with buyer age. The cost recursion therefore implies that $c^{SP}(a)$ is nonincreasing. Since $\sigma > 1$ and $D^{SP}(a)$ is nondecreasing, $R^{SP}(a)$ is nondecreasing. When $\Omega > 0$, the proportional current-return loss $\Omega/[NR^{SP}(a)]$ is consequently nonincreasing with age. This comparison holds the network and the common matching-stock term fixed.

Transmission to dynamic innovation values. Write $g(a)$ for the current-return term, equal to $\pi_\theta(a)$ in the decentralized value equation and $R^{SP}(a) - \Omega/N$ in the planner equation. Let V and w_H denote the value and R&D wage in the corresponding allocation. Consider a differentiable change indexed by τ along interior stationary value problems, holding R&D technologies fixed. Differentiating the value equations (14) and (23) after dividing by the corresponding R&D wage, and applying the innovation first-order condition, eliminates the direct policy derivative. Integrating the resulting linear equation under the corresponding

transversality condition and convergence of the integral gives

$$\partial_\tau \left(\frac{V(a)}{w_H} \right) = \int_a^\infty \exp \left\{ - \int_a^s [\rho + d - \lambda(u)] du \right\} \partial_\tau \left(\frac{g(s)}{w_H} \right) ds.$$

The positive weights capitalize current-return changes over future firm ages, including the subsequent network accumulation of entrants. Entry and incumbent R&D respond to the normalized values at age zero and at the incumbent's age, respectively. The age ordering of current-return ratios identifies the valuation force; the stationary comparisons in Section 5.2.2 also incorporate the induced changes in the network and the R&D wage.

Full surplus appropriation. At $\theta = 1$, the proportionality factor β/σ in (53) is common across ages. To establish its allocation implication, consider the auxiliary planner conditions obtained by omitting only $-\Omega/N$ from (23). Denote their product-line value and R&D multiplier by $\tilde{V}$ and $\tilde{w}_H$. The private value equation at $\theta = 1$ is then obtained by setting

$$V_1(a) = \frac{\beta}{\sigma} \tilde{V}(a), \quad w_{H,1} = \frac{\beta}{\sigma} \tilde{w}_H.$$

Consequently $V_1(a)/w_{H,1} = \tilde{V}(a)/\tilde{w}_H$ at every age, including entry at age zero. The incumbent innovation and entry conditions therefore give the same allocation. Since both systems retain the same resource constraint and state laws, this equivalence also holds at their stationary fixed point. The correction aligns relative R&D incentives without equating private and social payoff levels or removing the final-consumption markup. The comparison with the full planner then adds the matching-stock term and resolves the allocation and network jointly.

Age-independent network exposure. When $\zeta = \delta_M \zeta_0$, (6) reduces to $m(a_s, a_b) = \zeta_0 f(a_s)/N$. Supplier and buyer exposure per product line, production costs, and current returns are then independent of firm age. The matching-stock term is also common across ages. On the regular interior stationary branch with finite product-line mass and bounded values, incumbent innovation is therefore constant. Write $\lambda_0 < d$ for this rate and E_0 for entry. The innovation and entry first-order conditions and the R&D resource constraint imply

$$\begin{aligned} h'_H(\lambda_0) &= H'_E(E_0), \\ H_E(E_0) + \frac{E_0}{d - \lambda_0} h_H(\lambda_0) &= 1. \end{aligned}$$

Both the decentralized equilibrium and the planner satisfy these conditions. Under the power R&D costs, the first equation makes E_0 strictly increasing in λ_0 , and the left-hand side of the second increases from zero to infinity as λ_0 rises from zero to d . Hence the two allocations have the same entry, innovation rate, and product-line density $f_0(a) = E_0 e^{-(d-\lambda_0)a}$. Age-independent private–social return gaps change the levels of values and the R&D wage but not the allocation of the fixed R&D resource.

F.10 Proof of the R&D Reallocation Proposition

This subsection proves Proposition 2. All undated objects are evaluated at the stationary decentralized equilibrium in the proposition. Assume bounded continuous innovation with $\lambda(a) \leq d - \varepsilon$ for some $\varepsilon > 0$ at sufficiently old ages, finite positive static solutions, and a locally differentiable static log-consumption functional in the relative state neighborhood constructed below. These conditions ensure integrable static envelopes and bounded value solutions.

A feasible temporary transfer. Let $\ell(a) = f(a)h_H(\lambda(a))$ denote incumbent R&D labor per unit of firm age. Choose a nonzero, continuous and piecewise smooth profile $v(a, t)$ with compact support at positive ages and within a finite time interval $[0, T]$. For a transfer from entry, take $v \geq 0$. For a transfer within incumbents, take $v = v_o - v_y$, where $v_o, v_y \geq 0$ have equal age integrals at each date and are supported in two compact age intervals, with every age receiving labor older than every age giving it up. Thus $\int v(a, t)da = 0$ in the second case. The profiles can vanish at the time endpoints to preserve initial and boundary compatibility. For sufficiently small $|\tau|$, prescribe

$$\begin{aligned}\ell_\tau(a, t) &= \ell(a) + \tau v(a, t), \\ H_E(E_\tau(t)) &= H_E(E) - \tau \int v(a, t)da, \\ \lambda_\tau(a, t) &= h_H^{-1}\left(\frac{\ell_\tau(a, t)}{f_\tau(a, t)}\right).\end{aligned}\tag{54}$$

Density and matching follow their original state laws from the common initial state, with $f_\tau(0, t) = E_\tau(t)$. Both labor inputs remain positive, and the R&D resource constraint holds exactly at every date. For $\tau > 0$, the first case moves labor from entry to incumbent expansion; the second moves labor from younger to older incumbents and leaves $E_\tau(t) = E$. After T , entry labor and incumbent labor at each age return to their baseline levels, while innovation rates continue to adjust to the changed density.

Compact age support bounds v/ℓ , so the density equation has a positive characteristic solution with relative deviations of order $|\tau|$ during the intervention. After T , the relative density $r_\tau(a, t) = f_\tau(a, t)/f(a)$ satisfies

$$(\partial_t + \partial_a)r_\tau(a, t) = r_\tau(a, t) \left[h_H^{-1} \left(\frac{h_H(\lambda(a))}{r_\tau(a, t)} \right) - \lambda(a) \right], \quad r_\tau(0, t) = 1.$$

The right-hand side moves relative density toward one, so deviations cannot expand along a characteristic and $\sup_{a,t} |f_\tau/f - 1| = O(|\tau|)$. Cohorts entering after T have baseline density, while remaining deviations move into its exponentially declining age tail. Hence $\|f_\tau(t) - f\|_1 \rightarrow 0$, with N_τ bounded away from zero. The matching law and boundaries in (45) likewise preserve positivity, give $\|k_\tau - k\|_\infty = O(|\tau|)$, and imply convergence to the baseline.

These bounds establish feasibility over the full horizon and keep consumption changes uniformly of order $|\tau|$, allowing differentiation of discounted welfare. The bounded density value and integrable matching costate, whose source is bounded by a constant times $f(a_s)f(a_b)$, make the discounted terminal terms vanish. The density tails also remove the infinite-age boundary terms, justifying the integration by parts below.

Marginal social values. Write $\delta f = \partial_\tau f_\tau|_{\tau=0}$ and $\delta E = \partial_\tau E_\tau|_{\tau=0}$. Linearizing (54) and the density law gives

$$\begin{aligned} (\partial_t + \partial_a)\delta f(a, t) &= \left[\lambda(a) - d - \frac{h_H(\lambda(a))}{h'_H(\lambda(a))} \right] \delta f(a, t) + \frac{v(a, t)}{h'_H(\lambda(a))}, \\ \delta f(0, t) = \delta E(t) &= -\frac{1}{H'_E(E)} \int v(a, t) da. \end{aligned} \quad (55)$$

The term $-h_H/h'_H$ captures the reduction in innovation per product line when a larger density shares a given amount of age-specific R&D labor.

Let $W(a)$ denote the normalized social value of a density perturbation under the prescribed labor allocation, including subsequent changes in product lines and trading relationships. This evaluates the specified path and differs from the optimized planner value $V^{SP}(a)$. Using the static envelopes and matching-stock derivative from Section F.8, the adjoint of (55) is

$$\left[\rho + d - \lambda(a) + \frac{h_H(\lambda(a))}{h'_H(\lambda(a))} \right] W(a) = W_a(a) + R^{SP}(a) - \frac{\Omega}{N}. \quad (56)$$

Here R^{SP} and Ω are evaluated at the decentralized baseline, using the common-state static equivalence established in Section F.9. The matching costate in (49) has a positive source, giving $\Omega > 0$ through (43), including the boundary contributions derived in Section F.8.

Let $U(\tau) = \int_0^\infty e^{-\rho t} \log Y_\tau(t) dt$. Integration by parts using (55), the matching costate, and the boundary conditions established above gives

$$U'(0) = \frac{1}{\sigma - 1} \int_0^\infty e^{-\rho t} \int_0^\infty \left[\frac{W(a)}{h'_H(\lambda(a))} - \frac{W(0)}{H'_E(E)} \right] v(a, t) da dt. \quad (57)$$

Welfare therefore rises when labor moves toward uses with a higher marginal social value. The bracket compares this value for incumbent expansion and entry. For transfers within incumbents, $\int v(a, t) da = 0$ cancels the entry term, leaving the comparison between the receiving and giving ages.

The sign of the welfare gain. It remains to show that the marginal social value per R&D worker rises with firm age. The private value equation and the incumbent first-order condition $V(a) = w_H h'_H(\lambda(a))$ imply

$$\left[\rho + d - \lambda(a) + \frac{h_H(\lambda(a))}{h'_H(\lambda(a))} \right] V(a) = V_a(a) + \pi_\theta(a). \quad (58)$$

Thus private and social values have the same age-discount operator. Define the ratio of net social to private current returns by

$$r(a) = \frac{R^{SP}(a) - \Omega/N}{\pi_\theta(a)} = \frac{R^{SP}(a)}{\pi_\theta(a)} - \frac{\Omega}{N\pi_\theta(a)}.$$

The age-profile results in Section F.9 imply that $\pi_\theta(a) > 0$ is strictly increasing and $R^{SP}(a)/\pi_\theta(a)$ is nondecreasing. Since $\Omega/N > 0$ is common across ages, $r(a)$ is strictly increasing.

The bounded solutions of (56) and (58) express both values as discounted integrals of their current returns. The common factor depending on the starting age cancels in their ratio, giving

$$\begin{aligned} \frac{W(a)}{V(a)} &= \frac{\int_a^\infty \omega(s) r(s) ds}{\int_a^\infty \omega(s) ds}, \\ \omega(s) &= \pi_\theta(s) \exp \left\{ - \int_0^s \left[\rho + d - \lambda(u) + \frac{h_H(\lambda(u))}{h'_H(\lambda(u))} \right] du \right\} > 0. \end{aligned}$$

The weights are common to all starting ages. Increasing age from a_y to a_o removes the interval $[a_y, a_o)$, where r is strictly lower than on the remaining tail. The weighted average therefore rises, so

$$\frac{W(a_o)}{V(a_o)} > \frac{W(a_y)}{V(a_y)}, \quad a_o > a_y \geq 0.$$

Private optimality and free entry give

$$\frac{W(a)}{h'_H(\lambda(a))} = w_H \frac{W(a)}{V(a)}, \quad \frac{W(0)}{H'_E(E)} = w_H \frac{W(0)}{V(0)}.$$

The marginal social value per R&D worker is therefore strictly increasing with incumbent age and exceeds the entry value at every positive age. For $v \geq 0$, (57) gives $U'(0) > 0$. For $v = v_o - v_y$, the entry term cancels, and the equal labor amounts and ordered age supports imply that the gain at older ages exceeds the loss at younger ages, again giving $U'(0) > 0$. Differentiability gives $U(\tau) > U(0)$ for all sufficiently small positive τ in either case, proving Proposition 2.

F.11 Separate and Joint Corrections

The four-economy comparison in Section 5.2.2 varies the supplier's surplus share θ and whether firms face a matching-stock levy. Let $b \in \{0, 1\}$ indicate whether the levy is applied. In private payoff units, the stationary product-line value equation is

$$(\rho + d - \lambda(a))V(a) = V_a(a) + \pi_\theta(a) - w_H h_H(\lambda(a)) - b \frac{\beta}{\sigma} \frac{\Omega}{N}, \quad (59)$$

where π_θ is given by (53). The congestion correction charges each active product line $(\beta/\sigma)\Omega/N$ per unit time and rebates aggregate proceeds lump-sum to households. Firms take the levy as given when choosing innovation and entry. At each stationary allocation, Ω is evaluated from (26), using the social match value in (24) at that allocation's own network, revenues, innovation rates and entry. The entry condition, R&D resource constraint and state laws are unchanged.

The baseline decentralized economy uses the baseline surplus share and $b = 0$. The appropriation-only correction sets $\theta = 1$ and $b = 0$, while the congestion-only correction retains the baseline surplus share and sets $b = 1$. When $\theta = 1$ and $b = 1$, multiplying (59) by σ/β gives the planner's product-line value equation. Values and the R&D wage scale together, so the resulting innovation and entry choices coincide with the planner allocation. The levy thus implements the planner's R&D allocation under full surplus appropriation;

the congestion-only experiment applies the same rule under baseline contracts.

To distinguish value and R&D responses from changes in the network environment, we also hold the product-line density, static revenues and matching-stock charge fixed at each of the four reference allocations and recompute values, innovation, entry and the R&D wage under all four correction combinations. The additional decline in the entry R&D share from joint correction, beyond the sum of the separate corrections, ranges from 0.78 to 1.35 percentage points across these fixed environments, compared with 2.56 percentage points when the stationary network and age distribution adjust. Complementarity therefore arises even before the network environment adjusts, and is larger in the fully re-equilibrated comparison. The fixed-environment calculations isolate value and R&D responses rather than describing additional stationary equilibria.

G Numerical Implementation

The numerical implementation evaluates stationary densities from the analytical formulas in Section D and solves the value equations on an age grid. Transitions combine the forward state laws with the static allocation at each date. Further implementation details and diagnostics are provided in the replication documentation.¹⁷

The value equations are solved with finite differences. Deterministic aging is discretized with a one-sided derivative that uses the continuation value at the next age grid point. Under our sign convention, this derivative is implemented as a forward difference in each age dimension. Below, D_a denotes the resulting one-sided finite-difference matrix, including the boundary row at the upper age grid point; on two-dimensional grids, D_{a_s} and D_{a_b} denote copies of this matrix applied along the supplier and buyer dimensions.

G.1 Stationary Solution Procedure

The stationary calculation separates common state equations from allocation-specific equilibrium conditions. The common state equations map a candidate policy pair (λ, E) into the product-line and matched-product densities. The decentralized equilibrium and the planner allocation differ in how current returns are valued, in their value equations, and in the policy updates used to determine (λ, E, w_H) .

Common state blocks. For any candidate (λ, E) , the product-line density $f(a)$ is computed from the KFE and entry boundary, (3) and (4), with firm mass given by (2). Given f ,

¹⁷See MatLab/Documentation/NUMERICAL_METHODS.md in the replication package.

we compute N and the matched-product density $m(a_s, a_b)$ from the stationary kernel (6). These state laws are common to the decentralized equilibrium and the planner; they are simply evaluated at the policy objects generated by the relevant allocation problem.

Decentralized stationary equilibrium. Given (f, m) , we solve the static contracting block in two stages. First, with the production wage fixed at $w = 1/\mu^F$, we iterate the unit-cost equation (7) to convergence. The resulting $c(a)$ determines final-sales revenue $R^F(a)$ and input-expenditure weights $s^M(a_s, a_b)$. Second, holding these objects fixed, we solve the linear recursion for variable production costs $C(a)$ in (9), with intermediate-sales revenue given by (8). This step uses fixed-point iteration or an iterative linear solver. Unit costs are therefore solved before variable production costs. Once C is obtained, we compute bilateral transfers and the private flow payoff $\pi(a)$ from (11)–(12). Both static solution stages are repeated at each outer iteration using the updated (f, m) . The private product-line value then solves the HJB in (14). The outer update uses the private incumbent R&D policy (15), the free-entry condition (17), and R&D labor clearing (16) to update (λ, E, w_H) .

Stationary planner allocation. The planner uses the same state laws, but recomputes the intratemporal allocation from the first-best static block (38)–(39). The dynamic block consists of the planner product-line value $V^{SP}(a)$ and match value $V^M(a_s, a_b)$, which satisfy (23) and (24). The planner policy update applies the incumbent R&D allocation condition in (19) together with the planner entry condition (51).

Fixed-point variables. In both stationary calculations, the outer fixed point is over (λ, E, w_H) . State densities, static allocations, and values are recomputed conditional on each policy guess.

G.2 Stationary Densities

We discretize a on an evenly spaced grid $\{a_i\}_{i=1}^{N_a}$ with grid spacing Δa and $a_1 = 0$. Age integrals use trapezoidal quadrature weights $\omega_1 = \omega_{N_a} = \Delta a/2$ and $\omega_i = \Delta a$ at interior nodes. We evaluate (28) by cumulative trapezoidal integration of its exponent. This imposes $f_1 = E$ and preserves nonnegativity under either allocation.

Given f and N , we evaluate the matched-product density pointwise from (30). The closed form imposes both entrant boundary conditions without a separate transport solve. The resulting density is used in the static allocation and value equations.

G.3 Private Value Function V

Given (f, m) and the static contracting objects, the private product-line value solves the HJB in (14). On the age grid, the implementation uses the one-sided age-derivative matrix D_a and solves the implicit linear system

$$[(\rho + d)I - \Lambda - D_a] V = \pi - w_H h_H(\lambda),$$

where $\Lambda = \text{diag}(\lambda(a_i))$, π is the vector of private flow payoffs, and $h_H(\lambda)$ is evaluated pointwise. Policy iteration updates λ and w_H from the private R&D FOC and the R&D resource constraint.

G.4 Match-Value Function V^M

This block computes the match value used in the aggregate matching-stock derivative. The planner match value $V^M(a_s, a_b)$ solves the two-dimensional HJB in (24). The implementation discretizes supplier and buyer age on the same grid, applies one-sided finite differences in both age dimensions, and solves the resulting sparse linear system

$$\mathcal{M}^{SP} V^M = b^M,$$

where the source vector has entries $b^M(a_s, a_b) = (1 - \beta)R^{SP}(a_b)s(a_s, a_b)$. The matrix $\mathcal{M}^{SP}$ contains the discount, exit, link-destruction, innovation, and age-drift terms from (24). This compact representation is used inside the planner policy iteration before computing the aggregate matching-stock source.

G.5 Planner Value and Policy Updates

After solving for V^M , we compute Ω and $-\Omega/N$ from (43)–(44). The planner product-line value then solves the discretized version of (50). With the same one-sided age-derivative matrix D_a ,

$$\mathcal{A}^{SP} \equiv (\rho + d) I - \Lambda^{SP} - D_a,$$

and the source vector

$$b_i^{SP} = R_i^{SP} - \frac{\Omega}{N} - w_H^{SP} h_H(\lambda_i^{SP}),$$

the stationary planner value update is

$$\mathcal{A}^{SP} V^{SP} = b^{SP}.$$

The outer policy iteration updates incumbent R&D, entry, and the R&D resource multiplier from

$$\lambda_i^{SP} = \left\{ \frac{\varphi V_i^{SP}}{\gamma w_H^{SP}} \right\}^{\frac{1}{\gamma-1}}, \quad E^{SP} = \left\{ \frac{\varphi_E V_1^{SP}}{\gamma_E w_H^{SP}} \right\}^{\frac{1}{\gamma_E-1}},$$

and

$$1 = H_E(E^{SP}) + \sum_i h_H(\lambda_i^{SP}) f_i \omega_i.$$

Here V_1^{SP} is the value at the age-zero grid point. The displayed entry update is the positive interior case of (52); at a binding lower bound, the KKT condition in (51) applies.

G.6 Finite-Horizon Transition Solver

The planner transition maximizes the discretized welfare objective subject to the forward state laws. We parameterize R&D labor shares to enforce the resource constraint and compute welfare gradients through the induced state path.

Grid, initial state, and feasible controls. Let $a_i = (i-1)\Delta a$, $i = 1, \dots, N_a$, and $t_n = n\Delta t$, $n = 0, \dots, N_T$, with $\Delta t = \Delta a$ and $T = N_T \Delta t$. Age integrals use trapezoidal weights ω_i , with half weights at the two endpoints. The initial interior product-line density, matching kernel, and firm mass are taken from the decentralized stationary allocation on this grid. The age-zero boundaries are controls and may change at the reform date.

At date n , $r_{E,n}$ is entry's share of R&D labor and $r_{n,i}$ is the labor mass assigned to incumbent age cell i , including its quadrature weight. We use logits $X_{n,j}$, with $j = 0$ for entry and $j = 1, \dots, N_a$ for incumbents, to set

$$r_{E,n} = \frac{e^{X_{n,0}}}{\sum_{j=0}^{N_a} e^{X_{n,j}}}, \quad r_{n,i} = \frac{e^{X_{n,i}}}{\sum_{j=0}^{N_a} e^{X_{n,j}}}.$$

Entry and innovation are recovered as

$$E_n = (\varphi_E r_{E,n})^{1/\gamma_E}, \quad \lambda_{n,i} = \left(\frac{\varphi r_{n,i}}{\omega_i f_{n,i}} \right)^{1/\gamma}.$$

These expressions enforce feasibility at every date:

$$H_E(E_n) + \sum_i \omega_i f_{n,i} h_H(\lambda_{n,i}) = r_{E,n} + \sum_i r_{n,i} = 1.$$

In particular, a change in entry automatically changes the labor available for incumbents. Incumbent innovation also responds to the current density at a given allocation of labor.

At each date, the boundaries satisfy $f_{n,1} = E_n$ and $k_{n,i,1} = k_{n,1,j} = \zeta_0/N_n$, where $N_n = \sum_i \omega_i f_{n,i}$ includes the updated entry boundary. Innovation and the static allocation are evaluated after imposing these conditions. The age-zero nodes retain their trapezoidal weights, while the initial interior network is held fixed.

Cohort transport at constant R&D labor. Within a step, hold the R&D labor density assigned to a cohort constant at $\ell_{n,i} = r_{n,i}/\omega_i$. Substituting its innovation rate into the continuous density equation gives, along that cohort,

$$\frac{df}{dt} = (\varphi \ell_{n,i})^{1/\gamma} f^{1-1/\gamma} - d f.$$

To integrate this equation, set $v = f^{1/\gamma}$ and $A = (\varphi \ell_{n,i})^{1/\gamma}$. Then

$$\frac{dv}{dt} = \frac{A}{\gamma} - \frac{d}{\gamma} v, \quad v(t + \Delta t) = e^{-d\Delta t/\gamma} v(t) + \frac{A}{d} (1 - e^{-d\Delta t/\gamma}).$$

Consequently the interior density advances by

$$f_{n+1,i+1} = \left[e^{-d\Delta t/\gamma} f_{n,i}^{1/\gamma} + \frac{(\varphi \ell_{n,i})^{1/\gamma}}{d} (1 - e^{-d\Delta t/\gamma}) \right]^\gamma, \quad i < N_n.$$

This update accounts for the change in innovation per product line as the cohort's density changes at a fixed allocation of labor. The next date's age-zero density is imposed from E_{n+1} rather than transported from an existing cohort; mass leaving the upper age boundary leaves the truncated grid.

With N_n held fixed within the same step, the normalized matching law integrates to

$$k_{n+1,i+1,j+1} = e^{-\delta_M \Delta t} k_{n,i,j} + \frac{\zeta}{N_n} \frac{1 - e^{-\delta_M \Delta t}}{\delta_M}. \quad (60)$$

The new boundaries are imposed at date $n + 1$, and $m_{n,i,j} = f_{n,i} k_{n,i,j}$. Firm mass is recorded separately using

$$N_{f,n+1} = e^{-\delta_F \Delta t} N_{f,n} + E_n \frac{1 - e^{-\delta_F \Delta t}}{\delta_F}.$$

These forward operations, the static block at each date, and the welfare quadrature in Section G.7 determine the value of any candidate control path.

Compact matching representation. The initial stationary kernel depends only on the younger of the two ages, and (60) preserves that structure. Write $k_{n,i,j} = \kappa_{n,\min(i,j)}$. The corresponding kernel matrix K_n acts on a vector v as

$$(K_n v)_i = \sum_{j \leq i} \kappa_{n,j} v_j + \kappa_{n,i} \sum_{j > i} v_j.$$

Cumulative sums therefore evaluate the static cost and revenue equations using one vector per date instead of a dense matrix. This is an exact representation of the discretized matching kernel.

Discrete welfare derivative. Write z_n for the incoming state before the date- n boundary operations and $F_n(z_n, X_n)$ for the composed forward map to z_{n+1} . Let Y_n denote consumption after the boundary and static calculations. The welfare quadrature can be written as $U_{\text{path}} = \sum_{n=0}^{N_T} \omega_n \log Y_n$, where ω_n includes the discounted piecewise-linear integration weights and, at the final date, the constant-consumption tail. The minimized objective is

$$\mathcal{J}(X) = \sum_{n=0}^{N_T} L_n(z_n, X_n), \quad L_n = -\rho \omega_n \log Y_n.$$

For the discrete state adjoint A_n , the reverse recursion is

$$\begin{aligned} A_{N_T} &= \partial_z L_{N_T}, \\ A_n &= \partial_z L_n + (\partial_z F_n)' A_{n+1}, \\ \nabla_{X_n} \mathcal{J} &= \partial_X L_n + (\partial_X F_n)' A_{n+1}, \quad n < N_T. \end{aligned}$$

The recursion differentiates the full forward map, including the dependence of innovation and matching boundaries on the endogenous density and aggregate stock, and the static welfare envelope.

Optimization and terminal treatment. We initialize labor shares from a coarser-grid solution and optimize them using a quasi-Newton method with the discrete welfare gradient.

Final-date labor shares are fixed at the planner's stationary allocation, computed consistently with the transition dynamics. The terminal state and consumption remain endogenous. The four-economy stationary comparison is obtained independently from the stationary equilibrium conditions.

Decentralized policy transitions. For the incumbent R&D subsidy, the same forward state laws and welfare calculation are combined with the private policy conditions in Appendix H.4. We propagate candidate policies forward, solve private value equations backward, and jointly update entry, incumbent innovation, and the R&D wage. Terminal private values come from the stationary equilibrium under the policy, while terminal policies are evaluated at the state reached along the transition.

G.7 Transition Welfare Computation

For any computed path, household utility is

$$U_{\text{path}} = \int_0^{\infty} e^{-\rho t} \log Y_t dt, \quad Y_t = \frac{1}{P^F(t)}.$$

We integrate the computed path using piecewise-linear interpolation of log consumption. Consumption is held constant beyond the terminal date T , giving the terminal tail

$$U_{\text{tail}} = \frac{e^{-\rho T}}{\rho} \log Y_T.$$

The reference is the stationary decentralized allocation, with $U_{DE} = \rho^{-1} \log Y^{DE}$. The consumption-equivalent gain is

$$\Delta_{\text{CE}} = \exp \left\{ \rho (U_{\text{path}} - U_{DE}) \right\} - 1,$$

which is the computation underlying (27).

G.8 Numerical Accuracy and Convergence

Because accumulated relationships sustain innovation at older firms, the computations allow a wide range of firm ages. Further extending that range has little effect on stationary entry or consumption. The transition welfare gains are also stable across finer grids and longer horizons, holding estimated parameters fixed. Small differences between the transition's stationary reference and the independently computed stationary comparisons reflect numerical approximation. Grid settings, convergence criteria, and detailed validation results are documented in the replication materials.

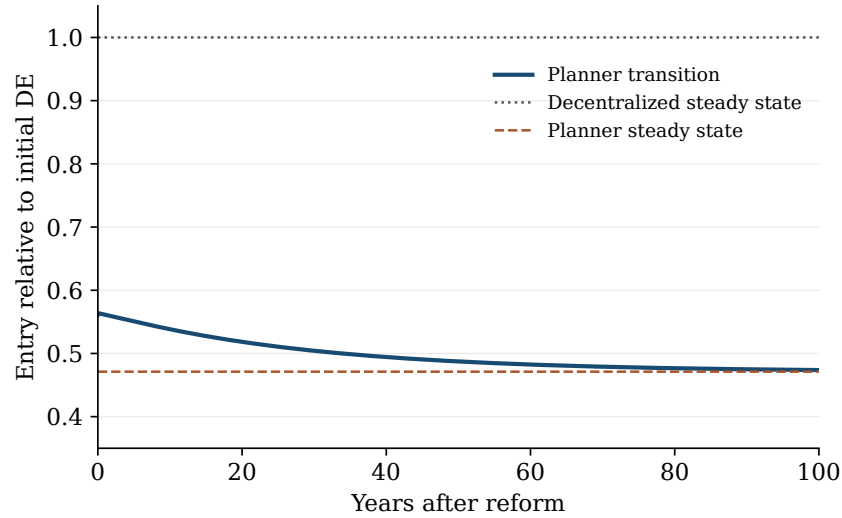


Figure A4: Entry Transition to the Social Planner Allocation

Notes: Entry is expressed relative to its initial decentralized stationary level. The solid curve is the planner transition. Horizontal lines indicate initial decentralized entry and the planner's stationary reference.

H Counterfactual and Policy Exercises

This appendix describes the planner entry path, the consumption paths under the planner and incumbent R&D subsidy, the feasible allocation-margin counterfactuals, and the uniform-subsidy exercise underlying the allocation and welfare results in Section 5.

H.1 Planner Entry Path

Figure A4 shows entry along the optimized planner transition over its first 100 years. Entry approaches its stationary level, while the age distribution continues to adjust.

H.2 Consumption Paths

Figure A5 compares consumption along the planner and optimal incumbent R&D subsidy transitions. Consumption gains build over time under both policies, with larger gains under the planner. The curves report consumption changes at each date, whereas the consumption-equivalent welfare gains summarize the full discounted paths.

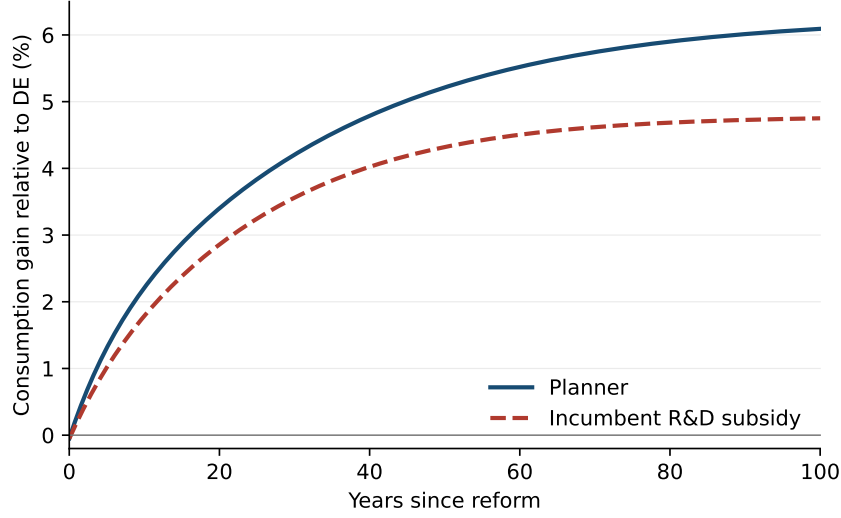


Figure A5: Consumption Gains along the Transition

Notes: Consumption is expressed as a percentage change from the decentralized steady state, $100[Y(t)/Y^{DE} - 1]$. Both paths start from the decentralized steady state. The solid curve shows the planner allocation; the dashed curve shows the optimal uniform incumbent R&D subsidy, financed by a lump-sum tax.

H.3 Welfare Decomposition by R&D Allocation Margin

For each date define the conditional incumbent R&D labor density

$$\psi_t(a) = \frac{h_H(\lambda_t(a))f_t(a)}{H_{H,t}}, \quad H_{H,t} = 1 - H_E(E_t), \quad \int \psi_t(a) da = 1.$$

Each hybrid selects the entry labor share and $\psi_t(a)$ from either the stationary decentralized allocation or the optimized planner path. The remaining incumbent labor is distributed using the selected conditional shares, and innovation rates are recovered from the current state density. All four paths are propagated from the same initial decentralized state using the cohort update in Section G.6.

To illustrate, the planner-entry/decentralized-age case sets E_t to the planner path and sets $\psi_t(a) = \psi^{DE}(a)$ at every date. Once the current density f_t is known, its labor allocation and innovation rate are

$$\ell_t(a) = [1 - H_E(E_t^{SP})]\psi^{DE}(a), \quad \lambda_t(a) = \left(\frac{\varphi \ell_t(a)}{f_t(a)} \right)^{1/\gamma}.$$

The implied policies advance the state while preserving total R&D labor. Entry can therefore change the incumbent population and its network. The other cases use the corresponding choices of entry and age shares.

Table A2: Welfare under Alternative R&D Allocations

Case	Entry shares	Incumbent-age shares	100× log CE
L_{00}	DE	DE	0.027
L_{10}	SP	DE	1.063
L_{01}	DE	SP	0.175
L_{11}	SP	SP	2.740

Notes: DE denotes the stationary decentralized labor allocation; SP denotes the optimized planner labor-share path. Each path starts from the same decentralized steady state. Log CE is relative to remaining at the stationary decentralized benchmark and multiplied by 100 for display.

Let L_{ij} denote log CE relative to stationary DE, where i selects planner entry shares and j selects planner conditional age shares. The Shapley contributions are

$$S_E = \frac{1}{2}[(L_{10} - L_{00}) + (L_{11} - L_{01})],$$

$$S_A = \frac{1}{2}[(L_{01} - L_{00}) + (L_{11} - L_{10})].$$

Table A2 reports the four underlying paths.

Thus $L_{11} = L_{00} + S_E + S_A$. The reference term L_{00} is kept separate because propagating fixed decentralized labor shares on the finite cohort grid differs slightly from remaining at the stationary DE benchmark. At a given allocation the baseline and first-best consumption prices coincide, since intermediate prices are efficient and the production wage offsets the final markup. These paths isolate R&D allocation margins; they are not decentralized equilibrium paths.

H.4 Uniform Subsidy to Incumbent R&D

Let $s_H \in [0, 1)$ be a constant subsidy rate on incumbent R&D expenditure. Entry R&D receives no subsidy, and all firms are eligible for support of product-line expansion after entry. Write $V_t^S(a)$ and $w_{H,t}^S$ for private product-line values and the R&D wage under the subsidy. The private value equation is

$$(\rho + d)V_t^S(a) = \pi_t(a) + \partial_t V_t^S(a) + \partial_a V_t^S(a) + \max_{\lambda \geq 0} \left[\lambda V_t^S(a) - (1 - s_H)w_{H,t}^S h_H(\lambda) \right].$$

The interior incumbent R&D and entry conditions are

$$V_t^S(a) = (1 - s_H)w_{H,t}^S h'_H(\lambda_t(a)), \quad V_t^S(0) = w_{H,t}^S H'_E(E_t).$$

The subsidy is financed by a lump-sum tax on the household equal to $s_H w_{H,t}^S H_{H,t}$, where

$$H_{H,t} = \int h_H(\lambda_t(a)) f_t(a) da, \quad H_E(E_t) + H_{H,t} = 1.$$

Fiscal transfers do not use research labor or change the physical resource constraint. We choose s_H to maximize consumption-equivalent welfare along the decentralized transition from the initial steady state. The optimal subsidy is 53.6%, yielding a 2.29% CE gain, or 82.5% of the planner gain in log CE units relative to stationary DE.

The same allocation can be implemented by a proportional tax on entry R&D. Let $\tau_E \geq 1$ be the entry-cost multiplier, and let $V_t^T(a)$ and $w_{H,t}^T$ denote values and the R&D wage under that tax. Incumbent R&D is unsubsidized, while entry satisfies

$$V_t^T(0) = \tau_E w_{H,t}^T H'_E(E_t).$$

Set

$$s_H = 1 - \frac{1}{\tau_E}, \quad w_{H,t}^S = \tau_E w_{H,t}^T, \quad V_t^S(a) = V_t^T(a).$$

Then $(1 - s_H)w_{H,t}^S = w_{H,t}^T$, so the incumbent value equation and both innovation and entry conditions coincide under the two policies. The state laws and physical resource constraints are also identical. Under the entry tax, revenue $(\tau_E - 1)w_{H,t}^T H_E(E_t)$ is rebated lump-sum. The household's R&D wage income net of fiscal payments is the same under the two policies, since

$$w_{H,t}^S - s_H w_{H,t}^S H_{H,t} = w_{H,t}^T + (\tau_E - 1)w_{H,t}^T H_E(E_t).$$

Thus the transformation preserves the entire transition allocation, consumption, and welfare, while changing the gross R&D wage and fiscal flows. The optimal subsidy corresponds to $\tau_E = 2.157$, or a 115.7% tax on entry R&D costs.